SPC-Eastwood-1

Machine Learning Classification of the Magnetosphere Structure at Lunar Distance

MSci Project Report

April 24th 2025

ys4920

Supervised by:
Professor Jonathan Eastwood

Assessed by:
Dr Robert J Forsyth

Word count: 9494

IMPERIAL

Lay Summary

Using Magnetic Fields and Machine Learning to Map Invisible Boundaries in Space

Beyond Earth's protective magnetic field lies a complex and dynamic environment shaped by the Sun's activity. Streams of charged particles (called plasma) from the Sun – known as solar wind – interact with Earth's extended magnetic field (called the magnetosphere), as shown in Fig. A. This interaction forms large-scale structures in space, such as the magnetotail (a long extension of the magnetic field on the side away from the Sun) and the bow shock (a boundary where the solar wind slows down quickly). These structures, and the boundaries between them, constantly shift in response to changes in solar activity. However, at the distance of the Moon, our understanding of this environment is still limited – even though it's important to future missions like NASA's Lunar Orbital Platform – Gateway, which will travel through this region.

This project sets out to classify the different regions of the Earth's magnetic environment (the magnetosphere) using a novel approach: by looking only at measurements of the magnetic field, instead of the more commonly used combination of magnetic field and plasma data. The goal was to build a tool that can quickly figure out which part of the magnetosphere a spacecraft is in, using minimal information. This type of tool is called a "now-casting" model, meaning it gives real-time updates like a GPS device but for space.

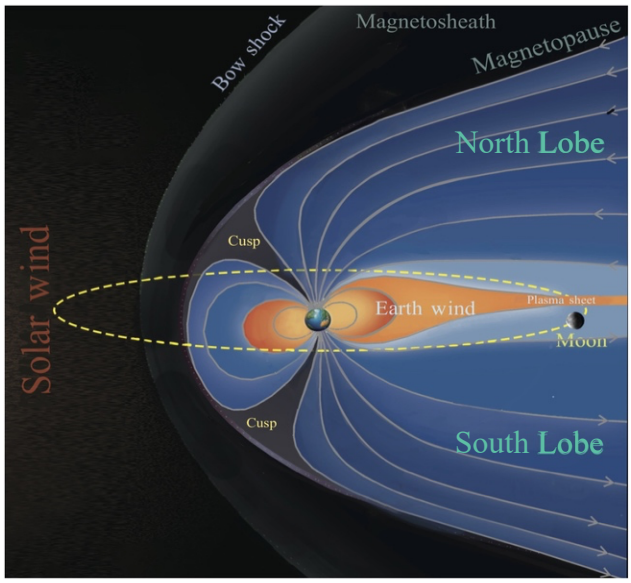


Figure A. A not-to-scale diagram of the magnetosphere and the Moon's orbit around the Earth. Source: [4].

To achieve this, fourteen years of data from the THEMIS B and C satellites (now called ARTEMIS P1 and P2) were analysed. First, a type of machine learning called unsupervised machine – specifically a method called *k*-means clustering – was used to identify and label where the satellite was, based on patterns in the speed and density of space particles (plasma), and the direction of the magnetic field. This method worked without needing any pre-labelled examples. Then, another type of machine learning, called supervised machine learning – using a model called XGBoost – was trained to recognise these regions using only magnetic field data. This model learned from labelled examples and achieved an overall accuracy of 96%, successfully distinguishing between space regions like the solar wind, and the north and south lobes of the magnetotail.

At the same time, the locations of important space boundaries – like the magnetopause (the outer edge of the magnetosphere) and the bow shock were tracked over time. The study also looked at how these boundaries changed during the Sun's 11-year activity cycle (the solar cycle). Although no strong patterns were found across the solar cycle, the methods used for classifying regions and tracking boundaries gave new insights into the large-scale structure of the Earth's magnetic environment at a lunar distance.

By showing that magnetic field data alone can be used to classify the near-lunar space environment, this project contributes a practical tool for future space missions. It also offers a scalable method that can help us better understand space weather and how it affects spacecrafts and their systems.

Abstract

Understanding space weather and the magnetosphere at lunar distance is critical for the safe operation of spacecraft such as NASA's Lunar Orbital Platform – Gateway. In this project, fourteen years of data from the THEMIS B & C (also known as ARTEMIS P1 & P2) satellites were analysed using two methods: an unsupervised machine learning technique (k-means clustering) model to classify regions within the magnetotail, and analysis of the power spectral density (PSD) of magnetic field data to identify bow shock (BS) crossings.

From this analysis, circular fits were produced to estimate the shape and positions of key magnetospheric boundaries at lunar distance. In geocentric solar magnetospheric (GSM) coordinates, the bow shock was found to have a radius of $46.4 \pm 9\ R_E$ and the magnetopause (MP) a radius of $26.6 \pm 7\ R_E$, both centred approximately at $(0,0)$. The average x-coordinates were $x = -37.09 \pm 10\ R_E$ for the BS and $x = -53.57 \pm 6\ R_E$ for the MP. A comparison across different years of the solar cycle revealed no significant trend in the positions of these boundaries.

Using these classifications, a supervised machine learning model (XGBoost) was trained to create a now-casting model for identifying magnetospheric regions using only magnetic field data. The model achieved an overall accuracy of $96.2 \pm 1.1\%$, with performance validated through confusion matrices, precision-recall analysis and cross-sectional plots of the magnetosphere. It successfully distinguished regions such as the solar wind and the north and south lobes of the magnetotail, indicating its potential for real-time region identification on Gateway. While the classification of the bow shock requires further refinement, the findings support the feasibility of using magnetic field data alone for operational space weather analysis at lunar distance.

Table of Contents

Lay Summary ii
Abstract iii
1 Introduction 1
1.1 Magnetosphere Structure 2
1.2 Lunar Orbital Platform - Gateway 4
1.3 THEMIS B & C (ARTEMIS P1 & P2) 5
1.4 Machine Learning 6
1.4.1 Unsupervised ML – k-Means 6
1.4.2 Supervised ML – XGBoost 8
1.5 The Solar Cycle 9
2 A Now-casting Magnetospheric Model 10
2.1 Overview 10
2.2 Importing Data Using PySPEDAS 11
2.3 The Magnetopause Boundary 12
2.3.1 Manual Classification 12
2.3.2 Automated Classification 13
2.4 The North and South Lobes 14
2.5 Magnetosphere Cross Section – Validation 15
2.6 Now-casting Supervised ML Model 16
2.7 Bow-shock Boundary 18
3 Further Investigation 21
3.1 Full Magnetosphere Crossing Plots 21
3.2 Yearly Magnetosphere Plots – Solar Cycle 25
4 Future Work 26
4.1 Two binary XGBoost Models 26
4.2 Sequential Nature of the Lunar Orbit 26
4.3 Storm Index 26
4.4 Aberration Correction 27
4.5 Now-casting the BS Boundary 27
4.6 Increased Resolution 27
4.7 Further investigation of $dPdyn/dt$ 27
4.8 Publication and Gateway 27
5 Conclusion 27
6 Acknowledgements 28
7 Bibliography 29

1 Introduction

As lunar exploration missions grow in ambition, understanding space weather becomes increasingly important. Space weather near the Moon consists of plasma and electromagnetic (EM) radiation from the Sun, which is shaped by Earth's magnetosphere. The magnetosphere is a large magnetic structure that surrounds the Earth and extends away from the Sun. The Moon passes through this structure and the different regions within during every orbit, making it an ideal location to study and improve our understanding of the magnetosphere.

The Lunar Orbital Platform – Gateway (Gateway) is a NASA-led space station designed as a multi-purpose lunar outpost. Serving as a docking station and habitation module, Gateway will support various scientific instruments and astronauts on extended missions in lunar orbit. Its unique near rectilinear halo orbit (NRHO) provides an opportunity to explore previously unobserved regions of the magnetosphere at lunar distance.

The magnetospheric regions Gateway will encounter include the solar wind (SW), the magnetosheath (MS), and the north lobe (NL) and the south lobe (SL) within the magnetotail (MT), as shown in Fig. 1. Distinguishing these regions and the boundaries between them is key to interpreting measurements from instruments onboard Gateway. The Blackett Laboratory at Imperial College London is contributing a magnetoresistive magnetometer (MAGIC) to the Gateway mission onboard the European Radiation Sensors Array (ERSA), one of Gateway's scientific payloads.

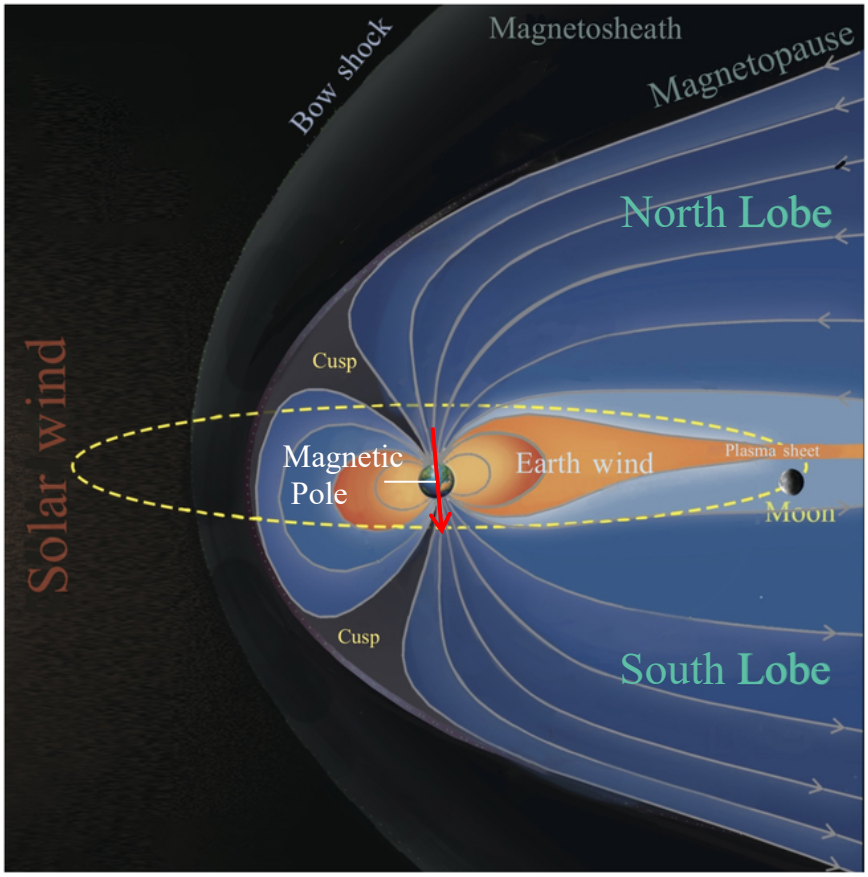


Figure 1. Schematic (not to scale) of Earth's magnetosphere, showing interaction with the solar wind and key regions such as the bow shock, magnetopause, lobes, and plasma sheet. Source: [1].

This project aims to build a now-casting model that distinguishes these magnetospheric regions using only magnetic field data, supporting the deployment of MAGIC on Gateway. To develop this model, we used 14 years of measurements taken by two currently operating lunar-orbiting satellites: Time History of Events and Macroscale Interactions During Substorms B & C (THEMIS B & C or ARTEMIS P1 & P2). We labelled the data by identifying distinct regions using unsupervised machine learning (k-means) and investigating the power spectral density (PSD) of the magnetic field. With this labelled dataset, we trained a supervised machine learning (ML) algorithm (XGBoost) to produce the now-casting model.

The remainder of this report is organised as follows**.** Section 1 provides essential background on the magnetosphere, the Gateway mission, the THEMIS satellites, relevant ML methods and the solar cycle. Section 2 outlines the development of the now-casting model, covering data import, preprocessing, and region classification. In Section 3, additional analyses are presented, including full magnetosphere crossings and variations across the solar cycle. Section 4 discusses directions for future work, including improvements to the model and its potential applications. The report concludes in Section 5 with a summary of the key findings and their implications for future lunar missions.

1.1 Magnetosphere Structure

The flow of all incoming solar plasma in the near-Earth space environment is shaped by the magnetosphere, hence the importance of understanding its general structure and origin. This magnetic field is generated by the flow of nickel and iron in the Earth's outer core through convection processes and the Earth's spin [2]. The dynamic (ram) pressure of the incoming solar wind and the opposing magnetic pressure from the magnetosphere reach an equilibrium across a boundary called the magnetopause (MP). The MP is compressed in the dayside and extends outward on the nightside forming a structure analogous to a windsock found in airports. This downwind extended region within the MP is the MT, which expands far beyond lunar distance. Due to the MP boundary, the plasma within the MT exhibits lower velocities ($|v| \approx 100\ \mathrm{km.\,s^{-1}}$). As a result of the Earth's magnetic poles, the magnetic field within the MT generally either points away from Earth, in the SL, or towards Earth, in the NL. To describe the direction of this magnetic field we use clock (θ_{clock}) and cone (θ_{cone}) angles. They are illustrated in Fig. 2 and are defined as follows

$$\theta_{\mathrm{clock}} = \arctan\frac{B_y}{B_z}, \qquad \text{where } -180^{\circ} < \theta_{\mathrm{clock}} < 180^{\circ}, \quad (1)$$

and

$$\theta_{\mathrm{cone}} = \arccos\frac{B_x}{|B|}, \qquad \text{where } 0^{\circ} < \theta_{\mathrm{cone}} < 180^{\circ}, \quad (2)$$

where B_x, B_y and B_z are the x, y and z components of the magnetic field in geocentric solar magnetospheric (GSM) coordinates respectively and $|B|$ is the magnitude of the magnetic field. In GSM coordinates, the origin is Earth's centre. The x-axis points from Earth to the Sun. The z-axis is defined as the component of Earth's magnetic dipole vector that is perpendicular to the Sun–Earth line and points roughly northward. Thus, the y-axis completes the right-handed coordinate system, pointing approximately in the direction of Earth's orbital velocity around the Sun.

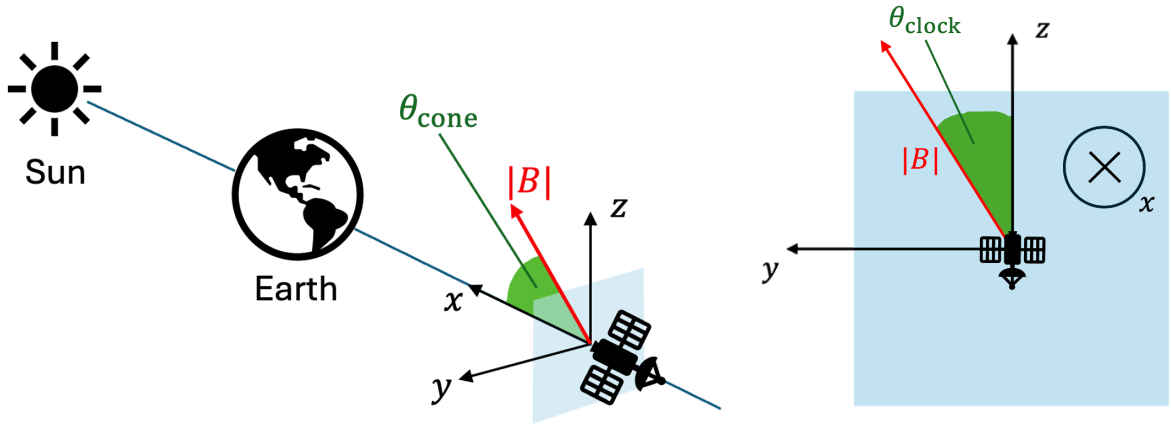


Figure 2. Schematic of the clock angle (θ_{clock}) and the cone angle (θ_{cone}) in geocentric solar magnetospheric coordinates (GSM), where θ_{clock} is shown in the $y-z$ plane projection. While GSM coordinates are centred at Earth's core, the origin here is placed at the satellite's position for visual clarity, assuming it lies along the x-axis.

The NL and SL exhibit very low plasma number densities ($n \approx 0.5\ \text{cm}^{-3}$). Between the lobes lies a boundary called the plasma (current) sheet, which contains denser and hotter plasma. This region is highly dynamic due to magnetic reconnection events that cause aurora on Earth. The plasma within the solar wind travels at velocities – relative to the magnetopause boundary – that far exceed the magnetosonic speed, which determines how fast disturbances can propagate through a plasma. As a result, information cannot travel upstream to divert the flow efficiently, causing it to abruptly change direction, speed, and/ or density, forming a bow shock (BS) – a thin region where physical quantities suddenly change. Between the BS and the MP, the plasma is redirected around and along the MP in the MS region. The plasma velocities ($|v| \approx 350\ \text{km.s}^{-1}$) and number densities ($n \approx 9\ \text{cm}^{-3}$) within the MS and SW regions are similar, making them difficult to distinguish. These two regions exhibit Kolmogorov turbulence, which describes the characteristic cascade of energy from large to small scales in fluid or plasma systems. Plotting the logarithm of the power spectral density (PSD) of the magnetic field against the logarithm of the frequency in these regions yields a linear relationship with a gradient $\alpha = -5/3$. In the BS, however, α is expected to deviate from this turbulence pattern, causing the PSD slope to flatten and become less negative. This deviation arises from high-frequency modes produced by plasma instabilities at the BS, driven by its interaction with the solar wind and the compressed, turbulent plasma of the MS. This interaction transfers energy into smaller spatial and temporal scales and flattens the PSD slope. Examples of these instabilities include mirror-mode and ion cyclotron waves [3], [4].

The current understanding of the magnetosphere at lunar distance is limited. Existing models of the magnetospheric boundaries and regions, such as Tsyganenko 2001 (T01) [5] and Shue 1997 [6], are accurate near the Earth but become inconsistent at greater (lunar) distances [7], [8]. One study identified MP crossings of THEMIS B & C over two years by visual inspection of magnetic and plasma data to produce an estimated circular MP model [9], which we compare to our values in Table III and Table V. Another study, with similar techniques, identified MP crossings of THEMIS B & C over five years of data [7]. By taking the outermost and innermost MP crossing within an MP crossing interval (due to spatial fluctuations of the boundary) and discarding the crossing in between, they were able to compare their results to MP radii predicted by the Shue 1997 model. Their measured innermost crossings differed by $-1.12 \pm 1.94\ R_E$ and their outermost crossings differed by $2.48 \pm 2.37\ R_E$. This study strengthens the

case for building an improved magnetospheric model at lunar distance. The implementation of unsupervised ML provides a unique opportunity to efficiently identify all available fourteen years of crossings whilst removing human bias.

1.2 Lunar Orbital Platform - Gateway

Given the limitations of current models at lunar distance, the upcoming Gateway mission offers a unique opportunity to gather new data and expand our understanding of the magnetosphere. Gateway (ARTEMIS IV) is the flagship of the ARTEMIS program, which aims to reestablish Moon exploration as a stepping stone to exploring Mars. The program is predicted to cost $93 billion. Astronauts on the Gateway are expected to operate for 30 to 90 days [10], [11]. Gateway will assume a 6.5-day NRHO around the moon [12], [13], [14], designed to enable constant communication with Earth and facilitate periodic proximity to the lunar north pole for lunar surface missions [13], [14], [15], as shown in Fig. 3. All existing lunar orbiting satellites have collected data at distances closer to the Moon along the Moon's equatorial plane [16]. The southern section of Gateway's NRHO will collect data in unexplored regions of the magnetosphere as the Moon orbits Earth, improving current models of its structure substantially.

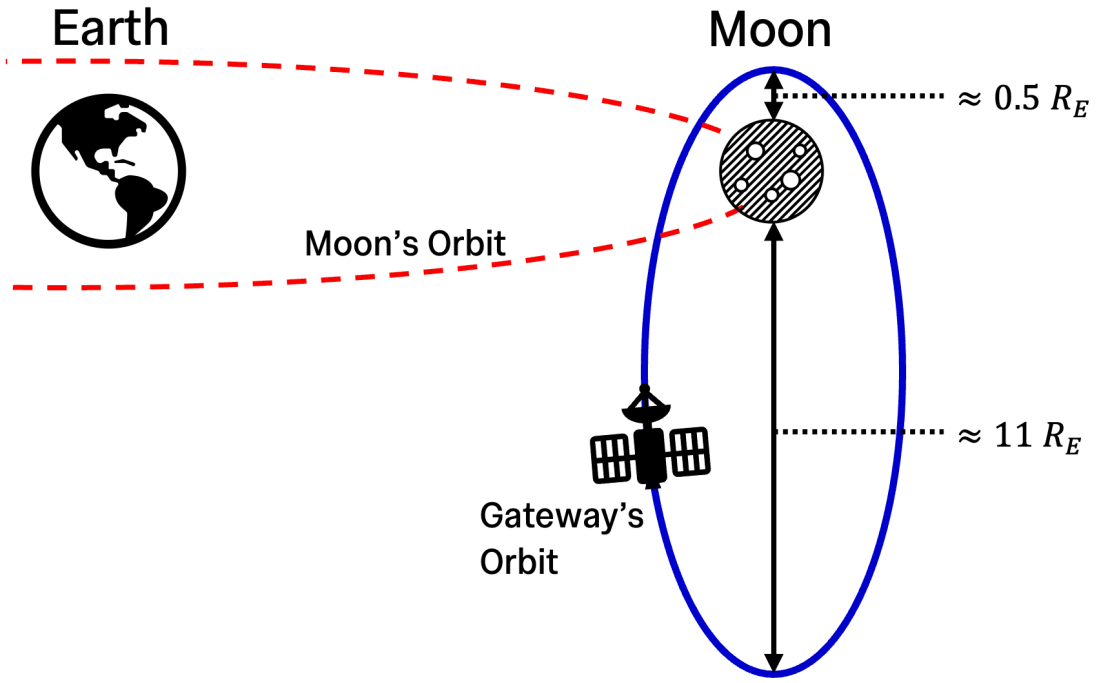


Figure 3. Schematic (not to scale) of Gateway's near-rectilinear halo orbit around the Moon, shown relative to the Moon's orbit around Earth. Distances are expressed in Earth radii (R_E), where 1 R_E = 6,378 km.

Imperial College London's contribution to ERSA involves the MAGIC device, a miniature magnetoresistive magnetometer weighing less than 80 grams. MAGIC is designed to measure the magnetic field and the spacecraft's attitude (orientation), crucial for determining the magnetospheric location of Gateway during transit and lunar orbit phases. It will also assist in interpreting data from other ERSA instruments. Equipped with both inboard and outboard 3-axis magnetic sensors, MAGIC can operate at varying sampling frequencies, either separately or in tandem [17]. The latest iteration of MAGIC was successfully deployed in the ESA's RadCube mission in 2017, where it demonstrated its capability to measure magnetic fields with an uncertainty of less than 1% [17], [18].

Gateway will also be equipped with comprehensive instrumentation such as the Heliophysics Environmental and Radiation Measurement Experiment Suite (HERMES) for measuring external magnetic fields, protons, electrons, ions, and lower-energy electrons. Notably, HERMES will also employ a magnetometer to cross-reference measurements with MAGIC, thereby eliminating anomalies and reducing measurement uncertainties [12], [19]. However, this project focuses on developing a now-casting model that uses only magnetic field data from MAGIC. This allows cross-validation across different instruments improving flexibility if plasma data is delayed, down-sampled, or otherwise unavailable to onboard decision-making systems.

1.3 THEMIS B & C (ARTEMIS P1 & P2)

To train and validate the now-casting model ahead of Gateway's deployment, we used historical data from the THEMIS B & C satellites, which have provided long-term measurements in lunar orbit. THEMIS B & C were originally launched in 2007 as an array of five satellites THEMIS A, B, C, D & E to investigate magnetic storms and aurora in Earth's orbit. In 2009, THEMIS B & C were reassigned and redirected to the Moon with a new name - Acceleration, Reconnection, Turbulence and Electrodynamics of the Moon's Interaction with the Sun P1 & P2 (ARTEMIS P1 & P2). Note that they are confusingly not part of NASA's ARTEMIS program discussed in section 1.2. In this report, we will adhere to the name THEMIS B & C to be consistent with the naming system in the Space Physics Environment

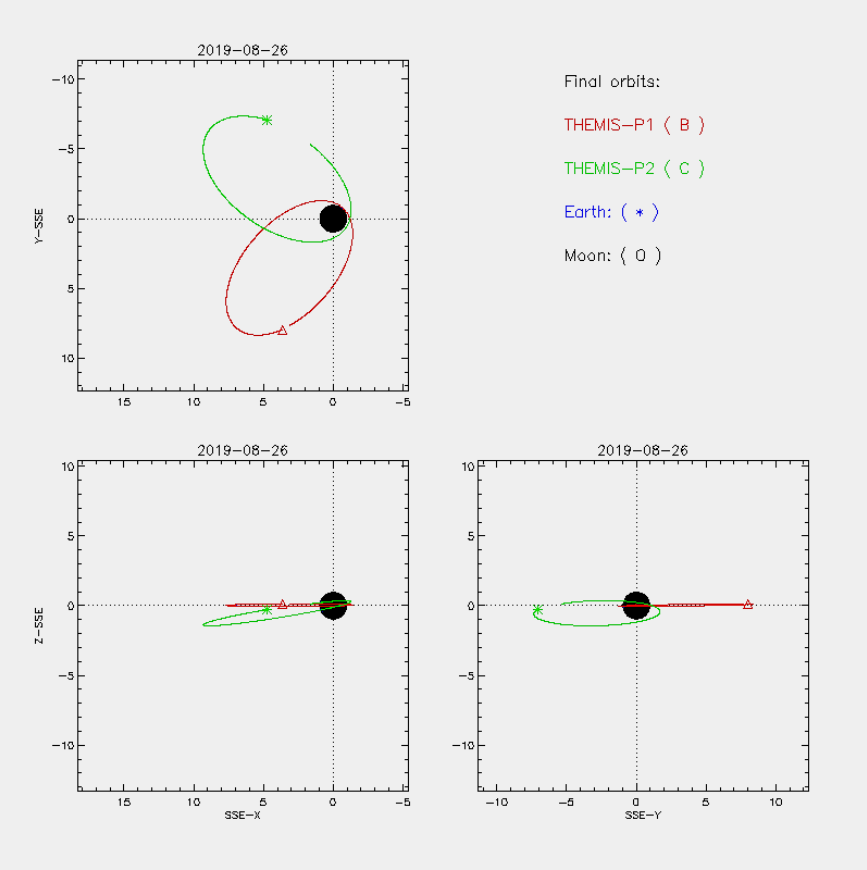


Figure 4. Orbits of the THEMIS B and C satellites (now ARTEMIS P1 and P2) around the Moon on 26 August 2019. Axes are in lunar radii, where $r_L \approx 1{,}737$ km. Coordinates are in the Selenocentric Solar Ecliptic (SSE) frame: the origin is at the Moon, the x-axis points toward the Sun, the z-axis is normal (northward) to the Earth's ecliptic plane, and the y-axis completes the right-handed set. Source: [20].

Data Analysis Software/ System Python package (PySPEDAS), which was used to import the data. In July 2011, THEMIS B & C entered equatorial, high-eccentricity lunar orbits. They orbit in opposite directions with an orbit of roughly 100 × 19,000 km altitude and a period of 26 hours [16]. Their orbits are relatively small on the magnetospheric scale; thus, we expect the plasma and magnetic measurements to be identical at hourly intervals.

These satellites have five scientific instruments on board. We imported data from one that measures the magnetic field - a FluxGate Magnetometer (FGM) - and another that measures plasma data – an ElectroStatic Analyser (ESA) [21] . These are the only active satellites with these measuring capabilities. Having accumulated fourteen years of magnetic and plasma measurements in Lunar orbit, these satellites offer a robust source of data for analysis.

1.4 Machine Learning

In the field of space weather analysis, when working with large datasets like those collected by the THEMIS satellites, ML techniques have emerged as transformative tools for both now-casting (real-time prediction of current conditions) and forecasting (anticipating future space weather events). These techniques are particularly advantageous in managing and analysing the vast amounts of data generated by decades of space missions [22]. As argued, the continued advancements in hardware and software technologies, coupled with the increasing availability of large, freely accessible datasets from both in situ and remote observations, have positioned space-weather research as an ideal candidate for the integration of ML [23].

All ML algorithms require the user to preset some parameters called hyperparameters. These parameters affect how the algorithm classifies the data and thus its overall performance. This will be discussed further in the following subsections.

1.4.1 Unsupervised ML – k-Means

Unsupervised clustering excels at identifying patterns in large, multi-dimensional (comprising many features) datasets with minimal prior knowledge or human input during training [22], [24]. In addition, it does not require labelled data. This approach eliminates human biases and "holds the promise of achieving true discovery or new insights" [22]. However, this advantage comes with challenges in terms of interpreting and validating the results. These issues can be mitigated by manually classifying a smaller subset of the data and quantifying the overlap with the unsupervised ML classification. Among unsupervised approaches, a commonly employed ML algorithm is k-means clustering [22], which can be used for now-casting and forecasting weather systems [25], [26]. This technique is suitable "to discover similarities in data, without deciding a priori the division between classes or, in other words, without specifying classes and their labels" [23]. Examples of the application of this technique in space weather research include the classification of sunspot groups [27] and the classification of solar wind into types based on different solar origins [28].

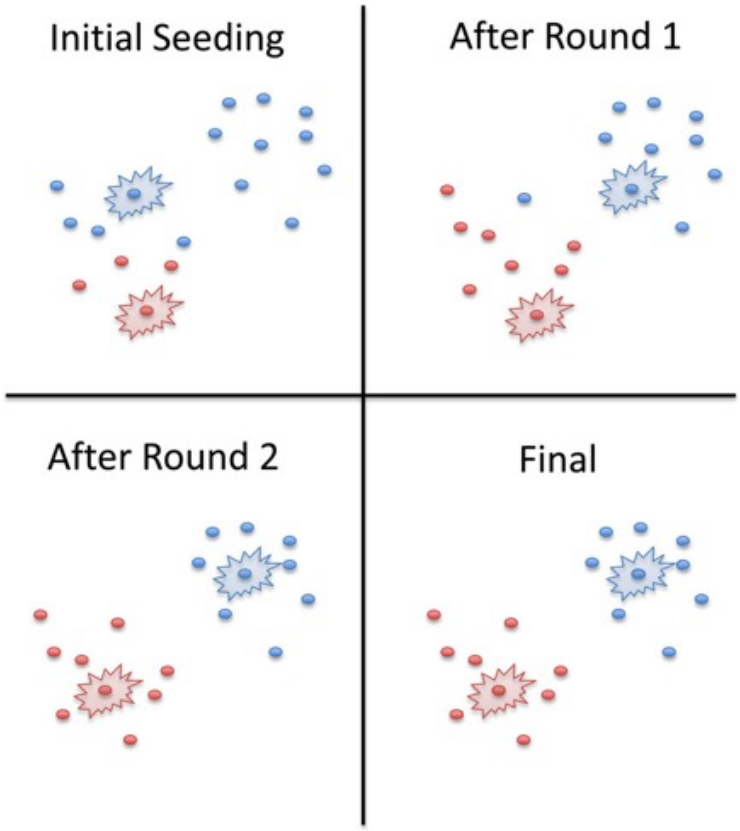


Figure 5. Illustration of the k-means clustering process in two-dimensional space, showing centroid updates and point reassignments across iterations [29].

The most important hyperparameter in the k-means algorithm is k, the number of clusters that the user would like the algorithm to identify. In this project, we only used a binary k-means algorithm, meaning $k = 2$. After setting the k number, the process is described as follows and is illustrated in Fig. 5:

1. The algorithm initially selects the centre of the clusters (centroids) in the N-dimensional space. The first centroid is chosen at random, and the second is positioned relatively far from the first.
2. Each data point is classified depending on the proximity to the two centroids in this N-dimensional space.
3. The position of each centroid is recalculated so that they are more central to the points classified to it.
4. Steps 2 and 3 are repeated until the centroids' location converges.

A trivial application of the binary k-means in a 2-dimensional space is shown in Fig. 10 (Section 2.3.2), where the features (dimensions) $|v|$ and n were used to distinguish measurements made inside and outside the MT. There is a large difference in the number of data points in each cluster. This allowed us to interpret the meaning of the clusters using our prior knowledge that the satellites spend a longer time outside the MT each full orbit around Earth. k-means was chosen for its speed, simplicity, and scalability across the large THEMIS B & C datasets [30]. While other methods like GMM or DBSCAN offer more flexibility, they were either too computationally intensive or required complex parameter tuning [31]. Therefore, k-means was the most practical choice for this application.

In section 2.3.2, we compare the k-means model (automated classification) against the hand labelled data (manual classification) by finding the overlap between the two datasets. This

overlap metric is the proportion of matching labels across the two datasets for each hour within the given time period.

1.4.2 Supervised ML – XGBoost

Supervised ML models rely on labelled data to learn a mapping from inputs (features like θ_{clock} and θ_{cone}) to known outputs. This allows them to be evaluated using metrics such as accuracy or RMSE. On the other hand, this also means that uncertainty in labels affects the model's performance. In contrast, unsupervised ML does not use labelled data and instead relies purely on patterns or clusters. The supervised ML algorithm used in this project was sci-kit-learn's

eXreme Gradient Boosting (XGBoost), a gradient-boosted decision tree (GBDT) ML library [33]. A GBDT is a combination of many low-accuracy decision trees, where a decision tree is a flowchart composed of nodes that aim to classify the data. As the model is trained on the data, each iteration of the XGBoost generates a new slightly improved decision tree. The ensemble of these low-level decision trees quickly reduces the uncertainty with low computational demand. A figure illustrating the decrease in error with the number of iterations is shown in Fig. 6.

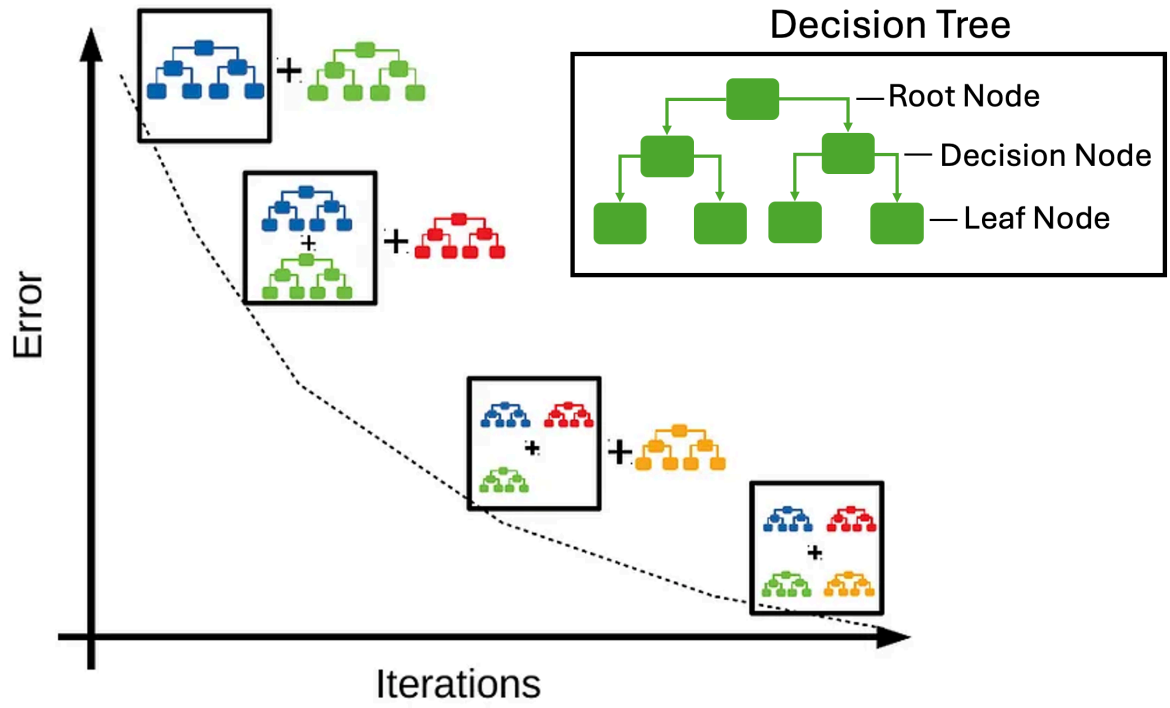


Figure 6. Illustration of how error decreases with each iteration in the XGBoost algorithm [32]. The inset shows the structure of a decision tree, including root, decision, and leaf nodes.

In comparison to other supervised ML algorithms, XGBoost excels due to its high performance on non-linear, high-dimensional data structure problems, and its speed and robustness to missing data [34]. These advantages are relevant for the project, given the 14 years of THEMIS B & C magnetic field data being used, which contains gaps and complex patterns.

Key hyperparameters include the learning rate, tree depth, and regularisation terms. It is important to tune these hyperparameters to avoid overfitting and improve generalisation. The model is trained on 80% of the data and tested on the other 20%. This enables analysis of performance by producing confusion matrices, precision-recall curves, feature importance

plots and the overall accuracy of the model. Confusion matrices visualise the false/ true positives/ negatives, as shown in Table. I.

TABLE I

		PREDICTED Classification	
	Classes	Predicted Negative (0)	Predicted Positive (1)
ACTUAL Classification	Actually Negative (0)	True Negative	False Positive
	Actually Positive (1)	False Negative	True Positive

Table I. Confusion matrix illustrating the performance of a binary classification model, showing the relationship between predicted and actual class labels.

We calculate the accuracy of each predicted classification class (a column in Table. I) by finding the proportion of true (correct) labels within the predicted classification class. The overall accuracy of the model is found by calculating the proportion of true labels across the entire dataset.

Precision-recall curves visualise the trade-off between precision, defined by

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}, \tag{3}$$

and recall (sensitivity), defined by

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}. \tag{4}$$

These curves are better suited for imbalanced datasets where the data is not evenly split which is ideal for this project as THEMIS B & C spend most of their transit in the SW region. The area under the curve (AUC) of a precision-recall curve summarises the performance of a model, ranging from 0 (poor model) to 1 (perfect model). The F1 score can be calculated by using the following formula

$$\text{F1 score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{5}$$

The F1 score is the harmonic mean of precision and recall, meaning that it is only high when both are high, punishing an imbalance [35].

1.5 The Solar Cycle

Due to the Sun's internal magnetic dynamo, the Sun's magnetic poles flip approximately every 11 years. This transition is termed a solar cycle. The solar dynamo is powered by differences in rotational speeds of plasma on the equator and the poles, and convective processes due to temperature differences [36]. During this cycle, the Sun exhibits changes in the number of sunspots, the solar irradiance, and the frequency of solar phenomena such as solar flares and coronal mass ejections (CMEs). Solar maximum refers to the period of peak activity, with high

sunspot counts and frequent solar eruptions, while solar minimum marks the quiet phase with fewer sunspots and reduced solar wind disturbances [37], [38]. The large change in plasma velocity and density caused by CME events increases the solar wind dynamic pressure (P_{dyn}). P_{dyn} is highly anti-correlated with the MP standoff distance, the distance of the sunwards of the MP boundary from the Earth's surface [39]. An increase in the P_{dyn} causes the compression of the magnetosphere as P_{dyn} reaches a new equilibrium with the opposing magnetic pressure along the MP [40]. The opposite occurs when there's a decrease in P_{dyn}. The solar wind dynamic pressure (or ram pressure) is given by

$$P_{dyn} = nm_p|v|^2 \,, \tag{5}$$

where m_p is the mass of a proton. This compression is made more extreme along the MT by the transverse (v_y and v_z) components of the solar wind, causing large distortions asymmetrically downwind [40]. Magnetohydrodynamic (MHD) simulations show that the MT at lunar distance is also affected by changes in P_{dyn} [40], distorting the BS and MP boundaries there [41]. The resulting fluctuations in these boundaries occur in orders of minutes and move by distances of approximately $\pm 15\ R_E$ for the MP [9]. Thus, we expect THEMIS B & C to cross the boundaries multiple times in short succession during more frequent and intense variations of P_{dyn}, dP_{dyn}/dt . CMEs cause large sudden changes in P_{dyn}, and during a solar maximum, CME events are more common. Therefore, we might expect to detect more crossings and less distinct boundary locations during a solar maximum and fewer, more defined crossings during a solar minimum. Multiple studies have investigated the relationship between P_{dyn} and the solar cycle, one of which found little to no correlation between them [39]. But there is a lack of research investigating if dP_{dyn}/dt is correlated with the solar cycle. In section 3.1 we investigate how the number of crossings changes throughout a solar cycle, but due to time constraints, we were not able to investigate dP_{dyn}/dt.

2 A Now-casting Magnetospheric Model

2.1 Overview

This project aimed to build a now-casting model that finds the magnetospheric location of a spacecraft using only magnetic field measurements. A simple approach to this problem would be to use rule-based heuristics, which set defined boundaries to classify data points. An application of this could be to classify a datapoint as an NL if $\theta_{\text{cone}} < 30^{\circ}$, or as an SL if $\theta_{\text{cone}} > 150^{\circ}$ based on our preexisting knowledge of these regions. Unfortunately, the physical measurements often deviate from expected behaviour. Examples include the dynamic interaction between the solar wind and the magnetosphere, instrument-induced noise, and high-frequency plasma wave turbulence, which introduces fluctuations that obscure large-scale magnetic field trends. In addition, these boundaries have a human bias associated with them, incorporating a systematic source of uncertainty that would be difficult to quantify. Unsupervised ML models are advantageous as they remove human bias, work well with large datasets with many dimensions, and distinguish patterns in data that would otherwise have been difficult for a human to decipher. A supervised ML model has the same advantages, with higher predictive accuracy, including the added benefit of enabling a qualitative evaluation of the performance. The largest disadvantage of a supervised model is its dependence on labelled data; thus, the accuracy of the labels is of utmost importance.

As an overview of our approach, we first classified/ labelled data from the in-situ THEMIS B & C satellites using all available plasma and magnetic field measurements. Secondly, we verify the classification using our existing knowledge of the physical phenomena of the magnetosphere. Lastly, we train a supervised ML model on magnetic field data classified in the previous steps. This approach minimises human bias, incorporates valuable plasma measurements without compromising on our project aim and allows qualitative analysis of the final now-casting model. This methodology was implemented for distinguishing the MP boundary and distinguishing NL and SL within the MT, shown in the lower portion of the flowchart in Fig. 7. These regions will be the focus of this report as this was what I focused on, as opposed to the BS boundary which was the focus of my partner.

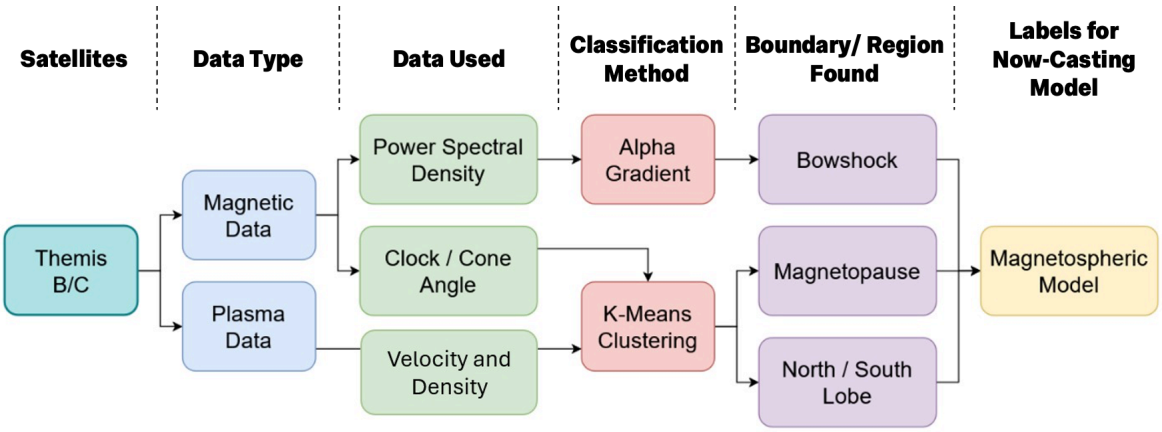
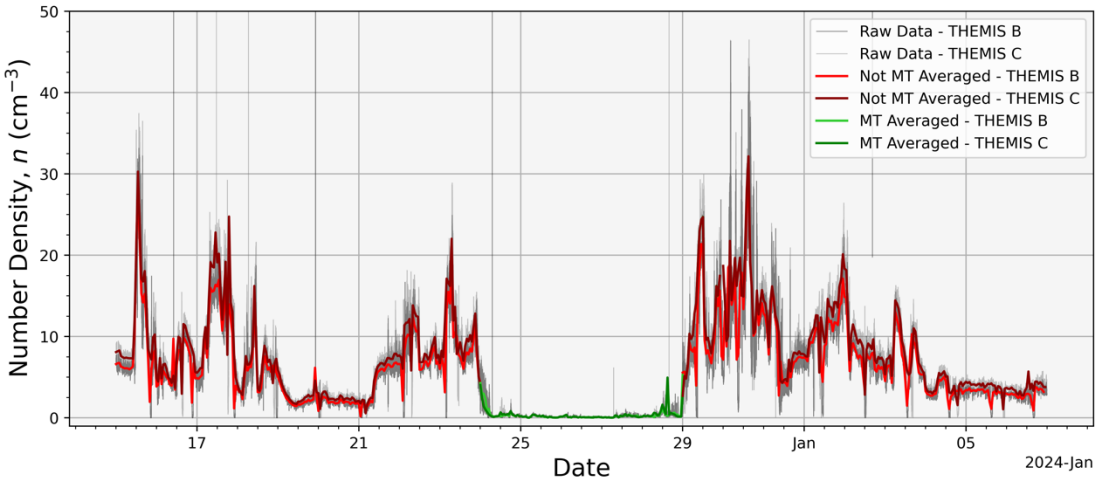


Figure 7. Flow chart summarising the methodology for generating the now-casting magnetospheric model, showing satellite data inputs, clustering method, and the identified regions used as model labels.

2.2 Importing Data Using PySPEDAS

All the data was imported using the PySPEDAS Python framework. We used THEMIS B & C data from July 2011, when the satellites entered a stable orbit around the moon. The data imported was in intervals of approximately 4.3 seconds concerning the period of the spin of THEMIS B & C. The imported data was calibrated into standard physical units and then converted into a CSV file. For example, to load magnetic data from the FGM instrument, we used the function `pyspedas.themis.fgm`. The data was then converted into hourly averages (shown in Fig. 8) to reduce sources of noise mentioned in section 2.1, which would help investigate the large-scale structure of the magnetosphere.



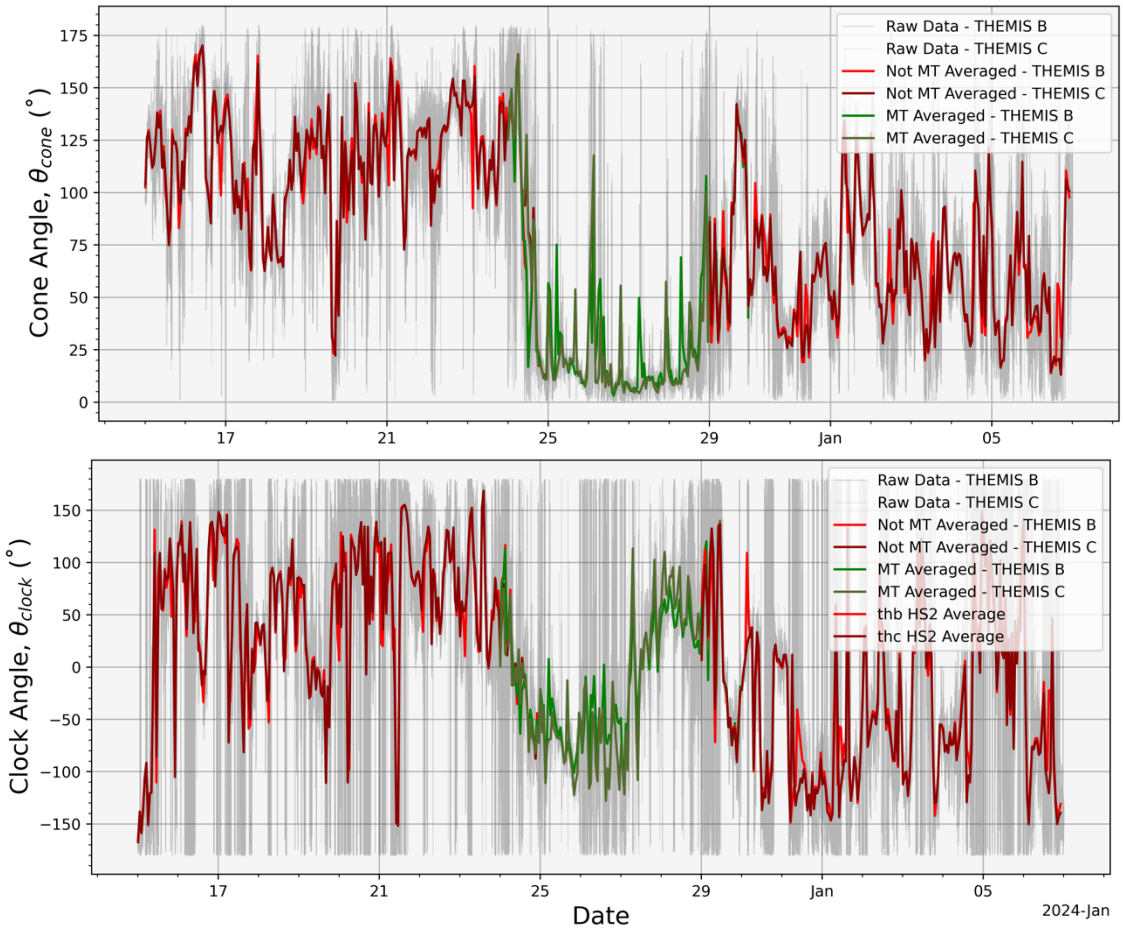


Figure 8. Time series plot of the number density, the clock angle and cone angle showing the hourly averaged intervals (coloured) in comparison to the raw imported data from THEMIS B & C. This shows data smoothing to reduce noise prior to further analysis and better represent the large structure of the magnetosphere. Manual classification methods in section 2.3.1 were used in these figures to classify data inside or outside the magnetotail (MT).

2.3 The Magnetopause Boundary

2.3.1 Manual Classification

To cross-validate our automated classification using k-means, we initially classified the MP boundary through visual inspection of time series plots of $|v|$, n and the ion energy flux of THEMIS B & C data from September 2023 to September 2024. The dates chosen were arbitrary. As mentioned in section 1.1, the plasma data is a clear indicator of the MP boundary, shown in Fig. 9. Due to the dynamic behaviour of the MP, some crossings were difficult to define as the spacecraft appeared to travel through the MP boundary multiple times within a few hours. The largest uncertainty in the MP boundaries due to these fluctuations was approximately ± 4 hours. Given the time-consuming nature of this work, the analysis was restricted to a single year of data.

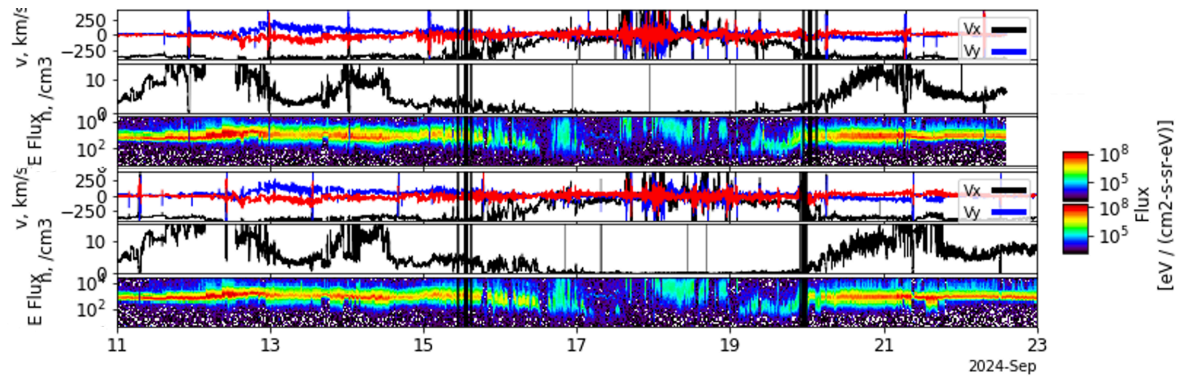


Figure 9. Time-series plots of THEMIS B (upper plots) and C (lower plots) passing through the magnetotail. Note that inside the magnetotail, the number density (n), the velocity magnitude ($|v|$) and the ion energy density decrease significantly.

2.3.2 Automated Classification

Using a binary k-means algorithm on $|v|$ and n data of one year of THEMIS B & C data, we clustered the data into either "inside the MT" or "outside the MT". The same timeframe as the manual classification was used here. We normalised the data before running the k-means algorithm to prevent feature bias. During this process, we compared the overlap with this automated classification to the manual classification as a validation metric. We ran a loop that ran this binary k-means algorithm using different combinations of data types (magnetic, plasma and/ or time measurements) and outputted the overlap metric. The highest overlap was found by using only $|v|$ and n data. A scatter plot of these two features is shown in Fig. 10. In this figure, we see that the $|v|$ is better at distinguishing the two clusters. Nonetheless, there is some overlap between the clusters, implying that relying on these two data types without any further processing is not enough to achieve a near-perfect accuracy.

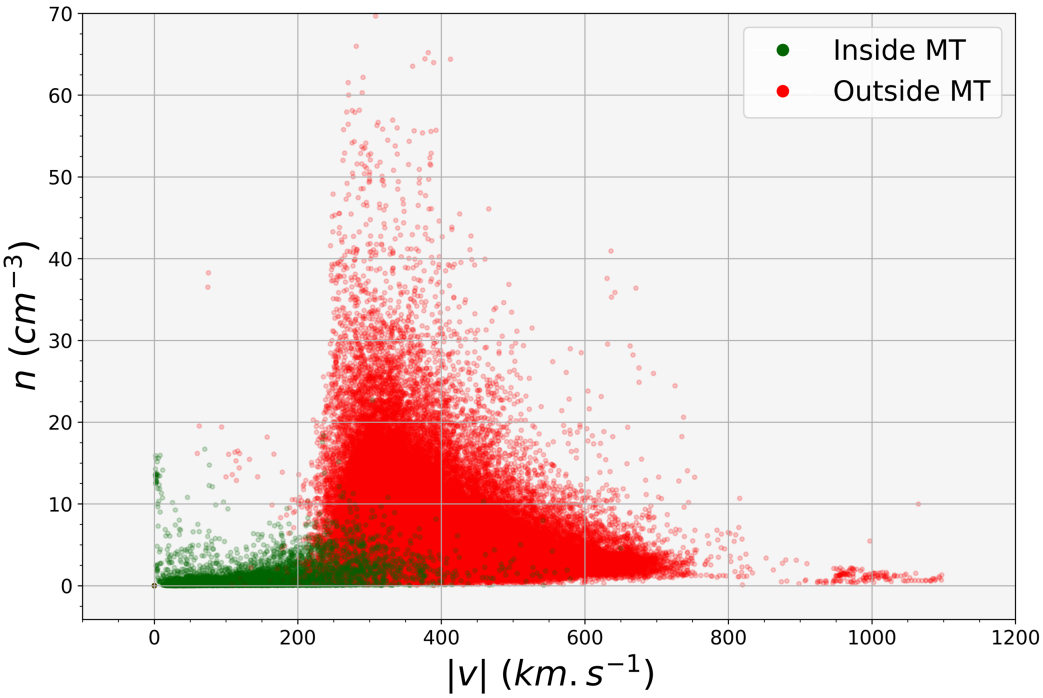


Figure 10. Scatter plot of the number density (n) against the velocity magnitude ($|v|$) across fourteen years of data classified by the k-means algorithm for distinguishing data inside or outside the magnetotail (MT).

To reduce spurious fluctuations in the clustering output, isolated data points were relabelled if surrounded by opposite-cluster labels for at least four hours (concerning the uncertainty in the manual classification) on either side, shown in Fig. 11. This smoothing step is justified by the Moon's gradual orbital motion, which makes rapid transitions between magnetospheric regions physically implausible. The correction ensures temporal continuity consistent with expected large-scale dynamics. In cases where the satellite crosses the magnetopause multiple times during a single interval - for instance, due to fluctuations in the boundary itself – such fluctuations are expected to occur on timescales much shorter than four hours. Therefore, this post-processing step is designed to preserve genuine crossings while removing isolated classification noise.



Figure 11. Visual representation of the post-processing technique used, where 0s and 1s represent the two classes.

This post-processing method increased the overlap by approximately 0.17%, resulting in a final overlap value of $98.8 \pm 1.1\%$ for THEMIS B & C. The uncertainty was found by taking the uncertainty in the manual labels and dividing it by the hours in the synodic month: $8/(24 \times 29.5) = 1.1\%$.

TABLE II

		AUTOMATED Classification	
	Classes	Outside MT	Inside MT
MANUAL Classification	Outside MT	7528	69
	Inside MT	33	1129
Overlap		99.6%	94.2%

Table II. The confusion matrix between the automated and manual classifications. The overall overlap was found to be $98.8 \pm 1.1\%$.

This high overlap gave us the confidence to extend the automated classification across all available fourteen years of data. Table II shows the confusion matrix between the automated and manual classification of the year of data. The lower accuracy in the automated classification of the data inside the MT can be explained by the lower proportion of data in the cluster, resulting in more false positives. Furthermore, data points where the cluster assignment changes are interpreted as MP crossings and are labelled accordingly.

2.4 The North and South Lobes

We used a second binary k-means algorithm on data previously classified as "inside the MT" on all fourteen years of THEMIS B & C data. Here data points were clustered into NL or SL. The lobes are defined by their magnetic field direction, thus we only used θ_{clock} and θ_{cone}

data. We expected a roughly even split due to the annual symmetry of the moon's location across the $x - y$ plane in GSM coordinates. Beyond this, we had no direct method of validating our results. Thus, we plotted the spacecraft's hourly location with the respective class attached to produce a magnetosphere cross-section as a visual validation method.

2.5 Magnetosphere Cross Section – Validation

We represented our automated classification process visually by plotting the hourly location of THEMIS B & C with its assigned cluster in GSM coordinates. We can compare this plot to our prior knowledge of the magnetospheric structure as a less rigorous validation method.

Using NASA's satellite situation centre web service (SSCWS) Python package, we imported hourly GSM coordinates for the THEMIS B & C. SSCWS is a client used to access NASA's SSCWeb application programming interface (API). An API is software that enables a user to request and receive data from an external service. Each cluster was assigned a distinct colour for visual interpretation, as shown in Fig. 12.

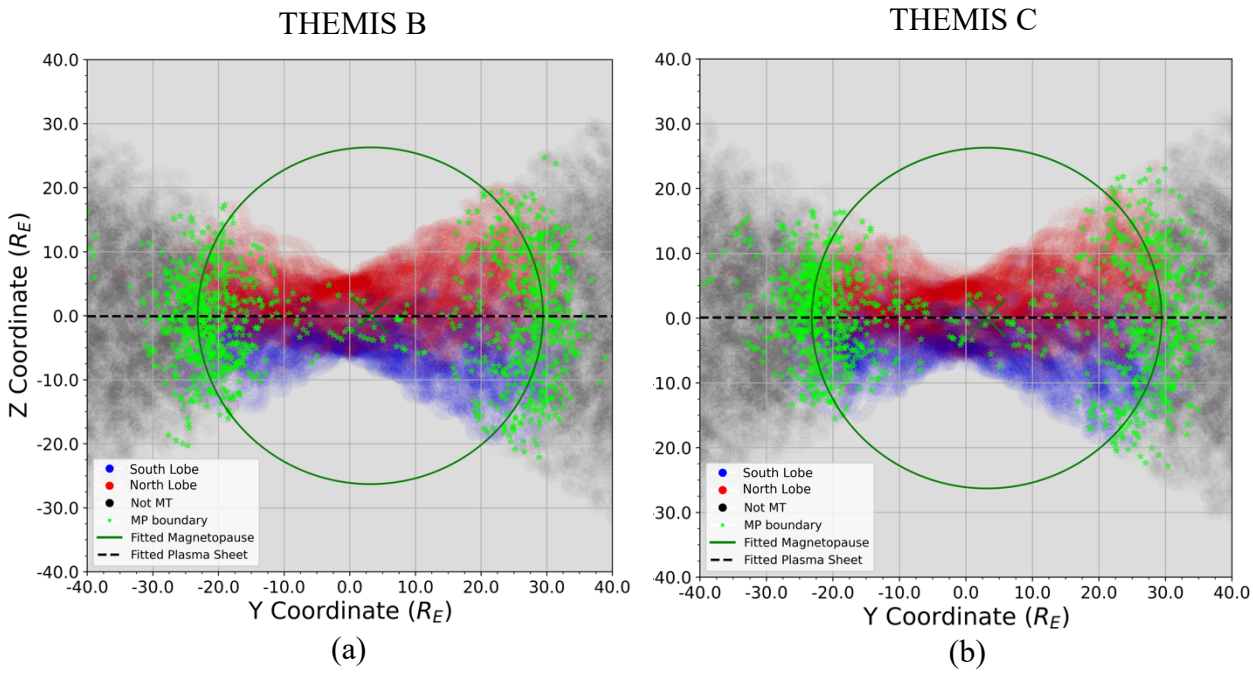


Figure 12. The hourly location of satellites near the magnetotail (MT) at lunar distance over a fourteen-year timescale. Each point was colour-labelled by two binary k-means algorithms described in sections 2.3.2 and 2.4. A circular fit of the magnetopause (MP) is shown. Only $x < 0$ data points are shown (downwind with respect to Earth).

The shape of the region covered by THEMIS B & C in Figure 12 resembles a "bow tie" shape centred roughly at (0,0). This can be explained by the 18.3˚ misalignment between the lunar orbital axis and Earth's magnetic pole. Most of the NL points are found in the positive z region and similarly for the SL points, which are in the negative z region. This is expected as the z axis in GSM coordinates is aligned with Earth's magnetic pole, shown in Fig. 1. A circular fit of the MP was plotted with the radii and centres of each THEMIS B & C shown in Table III. A large relative uncertainty of approximately 33% is expected due to the fluctuations caused by the dynamic solar wind interacting with the magnetosphere. In Table III, we compare a value found in the literature that falls within the bounds of uncertainty of the fit. Both centres of the fitted circles are off-shifted in the positive y and z direction. In conclusion, given the

large uncertainty in the radii, we don't expect the origin to be at exactly $(0,0)$. This off-shift will be expanded on in section 3.2.

TABLE III		
	Radius (R_E)	Centre (y, z) (R_E)
THEMIS B	27 ± 8	(1.23, 3.21)
THEMIS C	27 ± 9	(2.22, 4.05)
Literature [9]	26.3	Assumed to be 0

Table III. Radii and centres of the fitted magnetopause and in Fig. 12 in comparison to a radius found in the literature.

An approximation for the plasma sheet was found by taking the average z coordinate of all the NL and SL points and plotting the line bisecting the two. Over 14 years we expect the plasma sheet to be in line with the $x - y$ plane which is shown in Fig. 12. In addition, there appears to be MP crossing anomalies that have not been removed in the post-processing techniques near $(0,0)$. This suggests that there is room for improvement in the algorithm but, on the other hand, this could be due to high-energy solar events (e.g. CMEs). In total, these plots produced 2384 MP crossings. If the MP boundary was static we would expect approximately 350 MP crossings (two crossings every full orbit for over fourteen years). This shows that the algorithm is sensitive to the changing behaviour of the plasma across the boundary and that the MP boundary fluctuates as the satellites traverse through it. Nonetheless, this visualisation helped validate our automated classification to integrate their labels for a now-casting supervised ML model.

2.6 Now-casting Supervised ML Model

We trained an XGBoost model on Unix Time, α, Kolmogorov $-$ Smirnov distance, θ_{clock} and θ_{cone} of fourteen years of THEMIS B & C data. The Kolmogorov $-$ Smirnov distance (K- S distance) is a goodness of fit metric of the PSD of the magnetic field data. The test train split was 80/20. We ran a grid search on the model, which is a loop that finds the combination of data types and hyperparameter values that results in the highest accuracy score. A grid search over 36 combinations balanced complexity and efficiency, using standard XGBoost ranges. Parameters included `max_depth` (4–20), `max_leaves` (10–50), `learning_rate` (0.01–0.2), and `n_estimators` (50–200), chosen to explore underfitting, overfitting, and convergence trade-offs while remaining computationally manageable. The hyperparameters with the highest accuracy of 97.4% were `n_estimators` = 200, `max_depth` = 10, `max_leaves` = 20 and `learning_rate` = 0.2.

As shown in the confusion matrix in Table IV, the data is dominated by SW data points. Thus, a higher proportion of NL and SL data points are incorrectly predicted as the SW, reducing the accuracy of correctly identifying NL and SL in comparison to correctly identifying the SW. Two consecutive binary XGBoost models (analogous to the two binary k-means models used to label the data) have the potential to fix this issue, as there are very few false predictions between NL and SL. The overall accuracy of the model is also dependent on the accuracy of the labels themselves, reducing the overall accuracy to $97.4\% \times (98.8 \pm 1.1)\% = 96.2 \pm 1.1\%$. This accuracy is an upper boundary as the $98.8 \pm 1.1\%$ value is not a precise measure of the accuracy of the automated classification of the labels used in the model. Firstly, this value is the overlap of the manual and automated (k-means) clustering, and the manual

TABLE III

		PREDICTED Classification		
	Classes	Solar Wind	North Lobe	South Lobe
ACTUAL Classification	Solar Wind	39401	102	110
	North Lobe	309	2815	183
	South Lobe	301	196	2676
Accuracy		98.5%	90.4%	90.1%

Table IV. Confusion matrix of the now-casting supervised model. This illustrates how well the model performs in distinguishing each class.

clustering method has associated errors, one of which is human bias which is not included in the $\pm 1.1\%$ uncertainty. Secondly, $98.8 \pm 1.1\%$ only represents the overlap of the different classification methods on the clustering of data on either side of the MP boundary, not considering the NL and SL regions. Furthermore, this overlap was taken over a single year of data, thus, we assumed that it was a good representation of the full fourteen years of data.

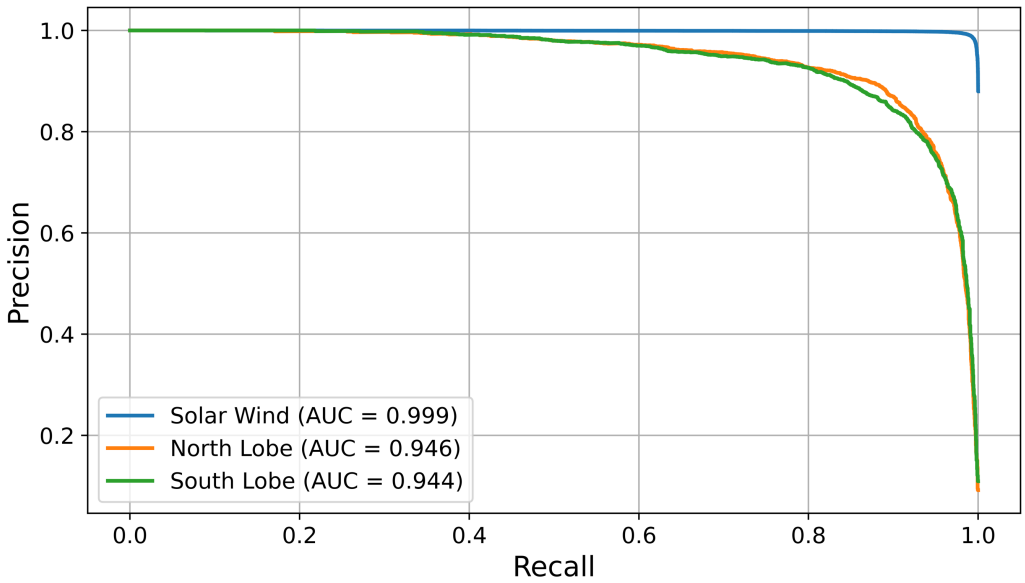


Figure 13. Precision–recall curves for the Solar Wind, North Lobe, and South Lobe classes, computed using a one-vs-rest approach. The area under the curve (AUC) is shown for each class, indicating high classification performance, particularly for the solar wind class (AUC = 0.999). Precision-recall curves were computed using the "class vs rest" method for each class.

As the model was trained on distinguishing three classes, the precision-recall curves were produced by individually comparing each class against the rest. As expected, the AUC is higher for the SW class, shown in Fig. 13, and the NL and SL classes have slightly lower values. The F1 score was 0.973, which shows that the model works relatively well.

Fig. 14 illustrates which feature is the most important for producing an accurate model by measuring which feature reduces the uncertainty the most. The sum of the feature importance scores is 1. The cone angle and α dominate, contributing to almost two-thirds of model accuracy. The cone angle is vital for distinguishing the NL from the SL and is also useful for distinguishing the two from the SW. The high importance of the α feature calculated from the PSD implies that fluid dynamics is different inside the MT in comparison to the SW, which is expected.

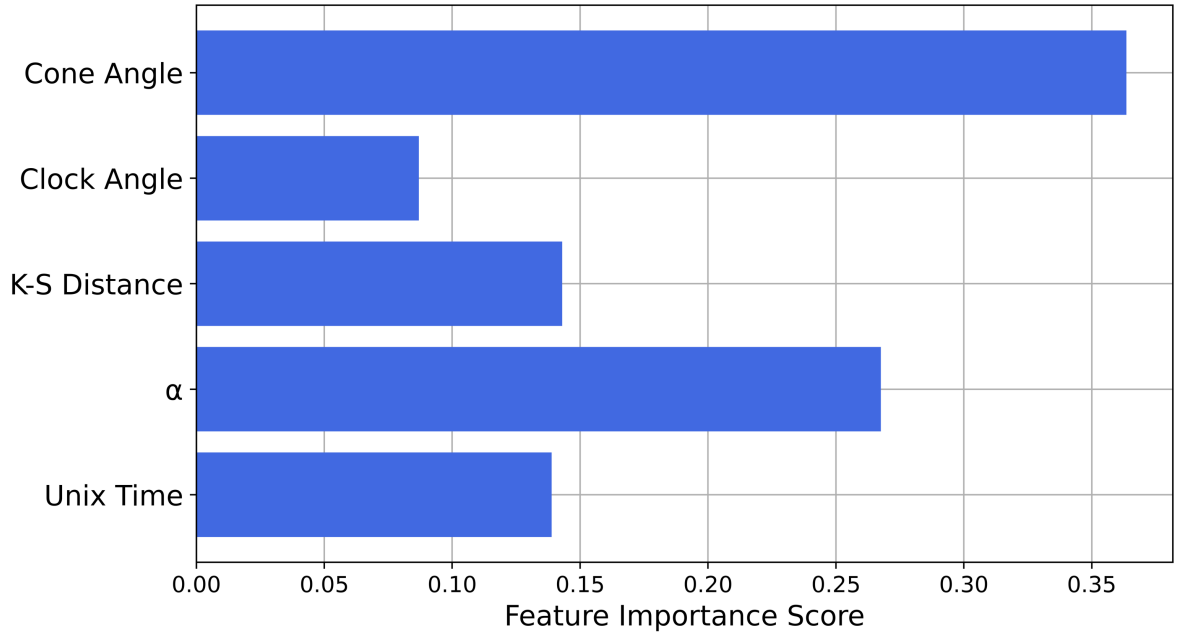


Figure 14. Feature importance scores for the five input features used in the XGBoost classification model. The cone angle and the spectral slope (α) contribute most significantly to the model's predictions, followed by the unix time, the Kolmogorov-Smirnov (K- S) distance, and the clock angle.

2.7 Bow-shock Boundary

We found the power spectral density (PSD) by using a Fourier transform of the x, y and z components of the magnetic field across intervals of six hours. The logarithm of the PSD was plotted against the logarithm of the frequency, and the average of the three components was

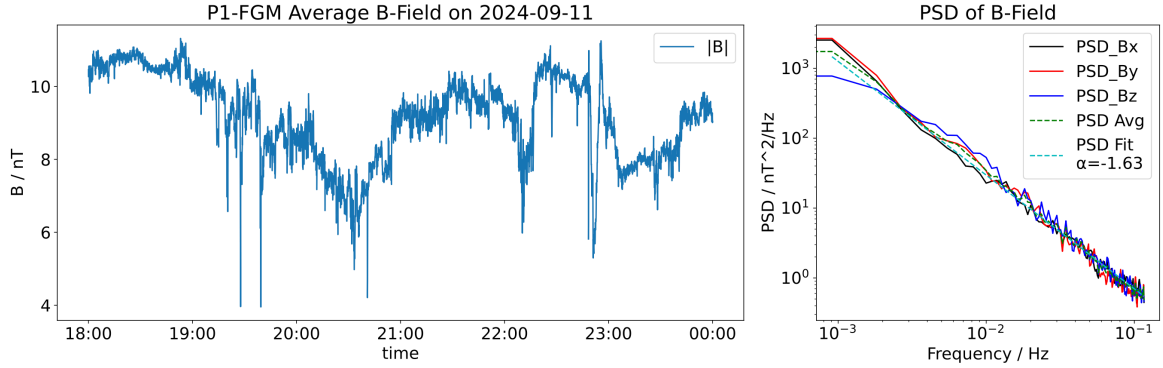


Figure 15. This interval exhibits Kolmogorov turbulence found in the SW and MS regions. On the left is the temporal plot of the magnitude of the magnetic field in a six-hour interval. On the right is the logarithm of the power spectral density of the magnetic field against the logarithm of the frequency with a linear fit (plotted over the average of the three spatial components) used to calculate the gradient α.

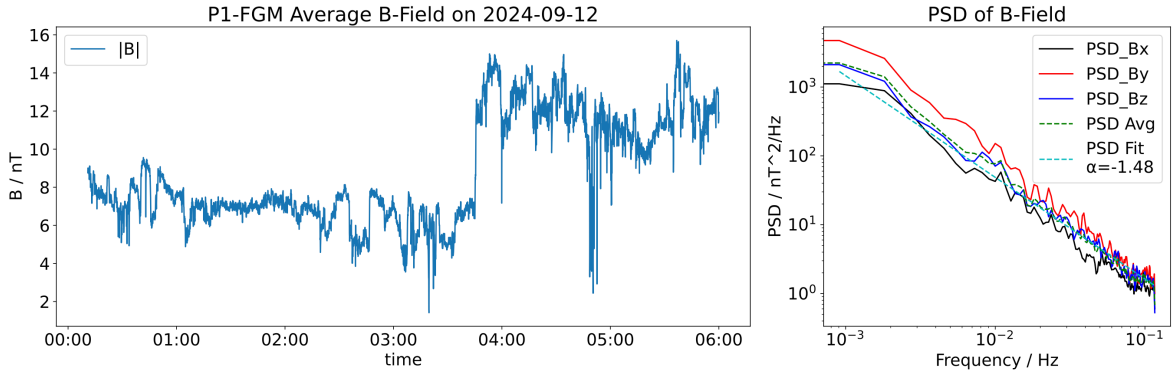


Figure 16. This interval exhibits non-Kolmogorov turbulence found in the BS boundary. On the left is the temporal plot of the magnitude of the magnetic field in a six-hour interval. On the right is the logarithm of the PSD against the logarithm of the frequency with a linear fit (plotted over the average of the three spatial components) used to calculate the gradient α.

analysed. As discussed in section 1.1, we expect an approximate gradient of $\alpha = -5/3$ in this plot except during a BS crossing. Fig. 15 shows Kolmogorov turbulence in the SW and MS and Fig. 16 shows the flattening of the gradient α in a BS boundary.

Values of α were found across all available data between 2011 and 2025 in six-hour intervals. To investigate how α varies during many lunar orbits, we generated a time-series plot of $\Delta\alpha = \alpha - \bar{\alpha}$, where $\bar{\alpha} = -1.60 \pm 0.01$ is the average α value in the SW. This highlighted intervals

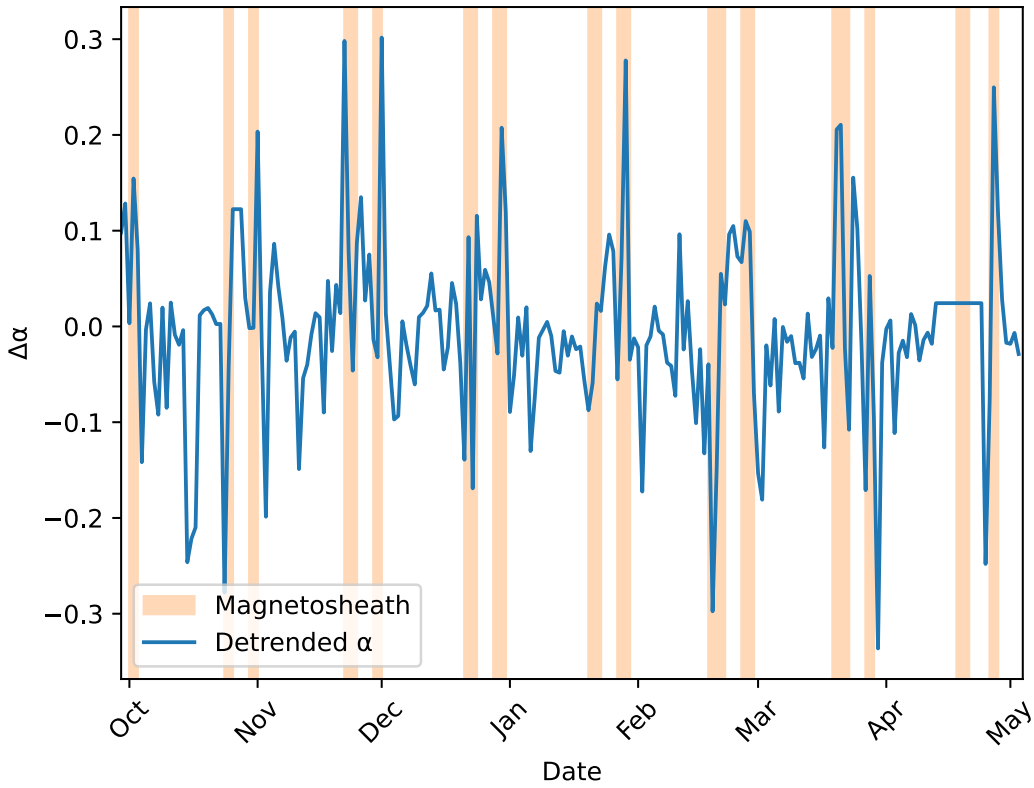


Figure 17. Time-series plots of $\Delta\alpha$ from the THEMIS B satellite, which shows the deviation of α from the solar wind average in until May 2024. The Magnetosheath regions are shown to highlight the periodic peaks at $\Delta\alpha \approx 0.15$ which we argue relates to the bow shock. The magnetosheath region was estimated by using the Berkeley magnetospheric model [42].

where the α varied from the average SW value. In Fig. 17, we see that every synodic month there are approximately two peaks at $\Delta\alpha \approx 0.15$.

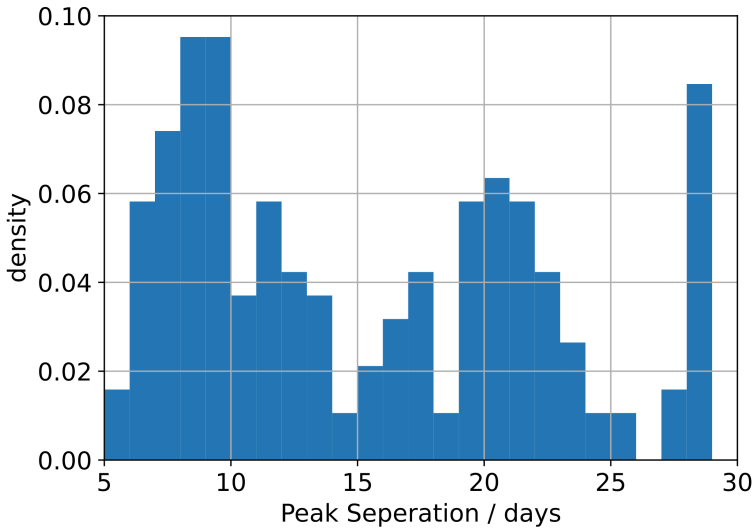


Figure 18. Histogram of the time between peaks of the gradient of $\Delta\alpha$ data. We see a general exponential decay distribution with two notable disturbances at approximately 8-10 days and 19-21 days.

By taking the gradient of $\Delta\alpha$, we highlighted these notable peaks and reduced the fluctuations of the smaller peaks near $\Delta\alpha = 0$. With the time series plot of the gradient of $\Delta\alpha$, we ran a `scipy.signal.find_peaks` function with a minimum peak height threshold of 0.1 and a minimum peak separation of 5 days. The minimum peak separation value was chosen given the satellites spend $\approx$ 4.3 days in the MT in each orbit. These values present a compromise between filtering out noise and capturing the peaks related to the BS. Then, we produced a histogram of the time intervals between peaks, as shown in Fig. 18. We focus on two notable disturbances, one at shorter intervals of approximately 8-10 days and the other at longer intervals of 19-21 days. We believe that these peaks are related to the long and short

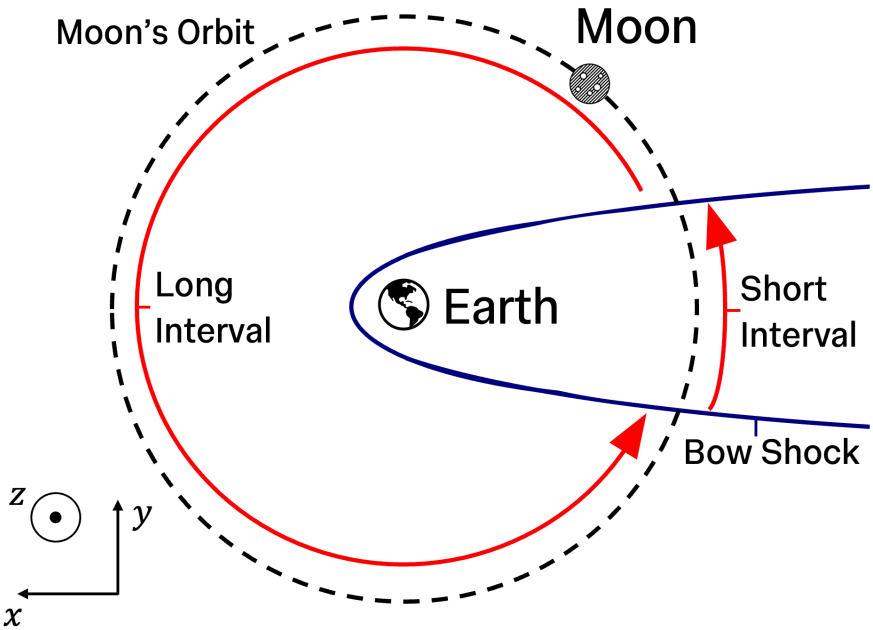


Figure 19. Illustration of the Moon's trajectory through the bow shock during its orbit around Earth. The diagram highlights the asymmetry in traversal time, showing a shorter interval on the nightside and a longer interval on the dayside due to the shape and orientation of the bow shock.

transit intervals with the BS boundary as illustrated in Fig. 19. The centres of these peaks add up to $(9 \pm 2) + (20 \pm 2) = 29 \pm 4$ days. Given the physical constraint of the periodic nature of the Moon's orbit, we expect to get a summation of 29.5 days (the synodic period), which falls within the bounds of uncertainty of value.

We evaluated each captured peak by applying a peak score to it to remove anomalies. The peak score is higher if nearby peaks differ by ≈ 11 and ≈ 19 days. This is shown in Fig. 20. The false positive peaks in Fig. 20 (purple vertical lines) are infrequent and trivial to isolate. This provided six-hour intervals where we expected the satellites to have crossed the BS by selecting the highest-scoring peaks. Attempting this technique at a higher resolution removed lower frequencies from the PSD plots, reducing the effectiveness of later methodology for finding the BS boundaries. Applying k-means to this dataset was not more effective as the data is only 2 dimensional, and the lunar orbit time constraint was difficult to implement. This could be explored further in the future. We attempted to implement this boundary into the now-casting model but found that it was very poor at predicting the BS boundary. 92% of the BS boundary data points were labelled as an MT or SW region. Due to time constraints, we were not able to develop the model further to improve the prediction of the BS boundary in the now-casting model.

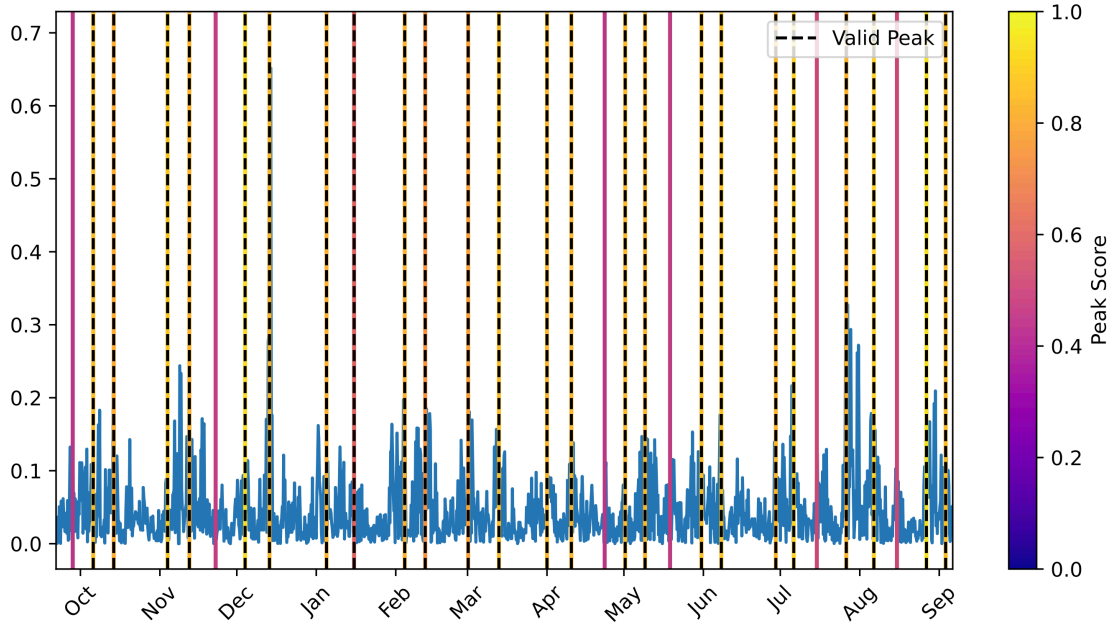


Figure 20. Time series of the absolute gradient of $\Delta\alpha$ from October 2022 to September 2023, with scored and validated peaks overlaid. The colour bar represents the peak score, and validated peaks (dashed lines) highlight the periodicity of lunar crossings - alternating between short (~11 days) and long (~19 days) intervals.

3 Further Investigation

3.1 Full Magnetosphere Crossing Plots

Combining the location of the BS and MP boundaries, we produced the plots shown in Fig. 21. These plots include crossings from both THEMIS B & C. As expected, the radius of the BS boundary fit is larger than the radius of the MP fit, with values shown in Table V. The radius of the MP fit falls within the bounds of uncertainty of the radius found in the literature

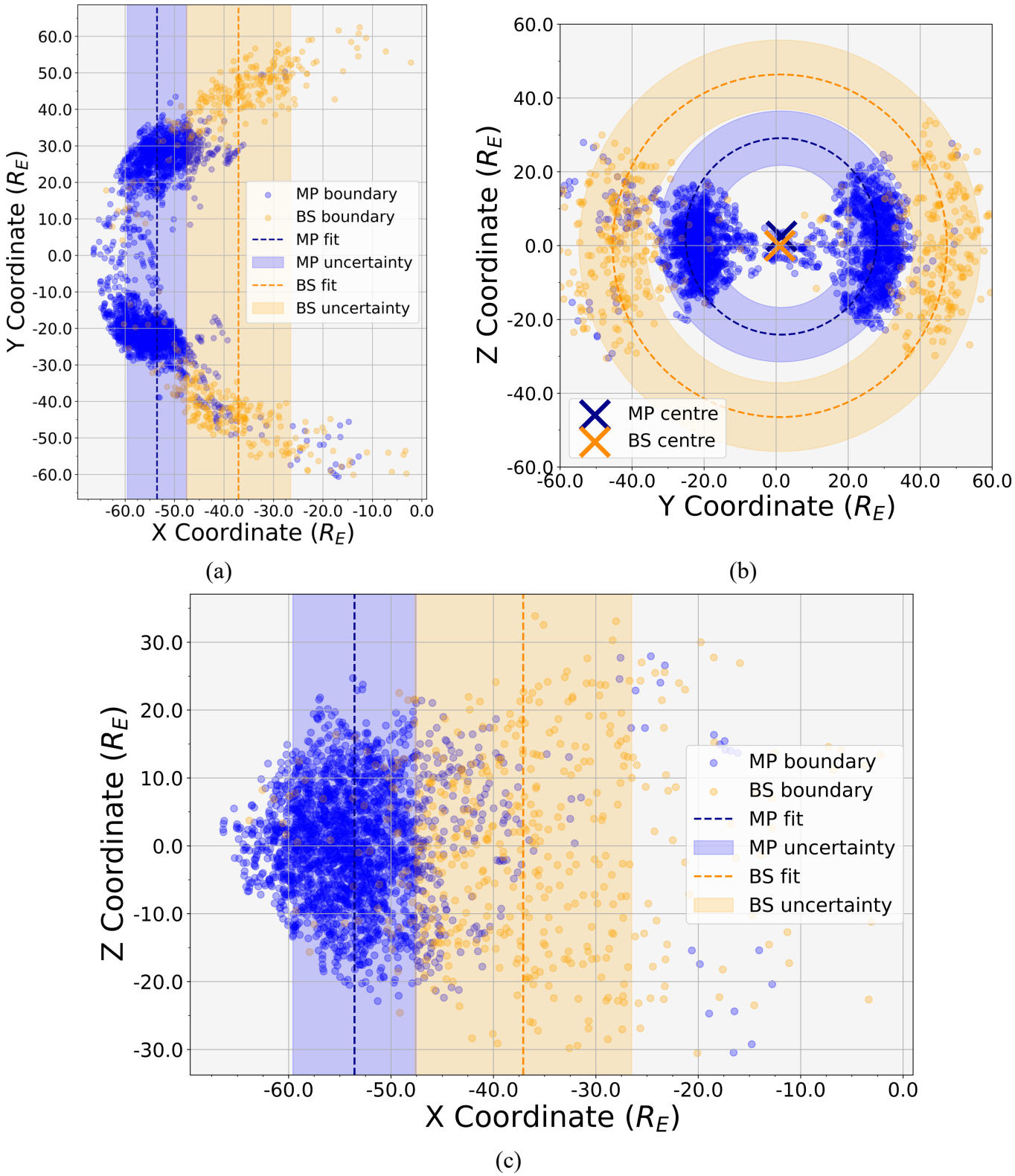


Figure 21. Magnetopause (MP) and bow shock (BS) boundary crossings from fourteen years of THEMIS B & C data, shown in hourly resolution across three geocentric solar magnetospheric planes: (a) $x-y$, (b) $y-z$, and (c) $x-z$. In the $y-z$ plane, circular fits were applied to determine the best-fit radius and centre for each boundary. In the $x-y$ and $x-z$ planes, vertical dashed lines represent the average x-position of the boundaries, with shaded regions indicating uncertainty.

[9], which was found by plotting a circular fit on 2 years of THEMIS data and assuming the centre is at the origin in the $y-z$ GSM plane. We also assumed a circular fit rather than a more general ellipsoidal fit as the data is limited to the transit of the THEMIS B & C satellites, with only $\approx 180^{\circ}$ of data available. The high uncertainty in the radius of both circular fits is likely due to the dynamic fluctuations of the magnetic field with the transient solar wind. In Fig. 21(a), we find that the BS is at $x = -37.09 \pm 10\ \mathrm{R_E}$ whereas the MP is further away at $x = -53.57 \pm 6\ \mathrm{R_E}$. This is purely due to only data being collected in the orbit of the Moon around the Earth. This is also shown in Fig. 21(c). As stated in section 2.5, many MP boundaries are being detected near $(0,0)$ in the $y-z$ plane, outside the bounds of uncertainty of the fitted circular boundary. This suggests either a misclassification in the k-means model or highly dynamic magnetospheric behaviour. In addition, there are far more MP crossings than BS crossings, which is discussed further in section 3.2. The minimum number of BS crossings was selected by choosing the crossings with the highest associated peak score. In the dusk side (negative y direction) there are notably more MP boundaries outside the bounds of uncertainty of the fit than in the dawn side (positive y direction). In addition, there were also approximately 20% more MP crossings on the dusk side.

Given our initial understanding of the magnetosphere at lunar distance we expected the boundaries to be centred near $(0,0)$ in the $y-z$ plane. As shown in Table IV, we find that the centres are off-shifted slightly with the MP and BS located at $(1.45, 2.50)$ and $(1.02, -0.04)$. Nonetheless, the high uncertainty caused by the dynamic fluctuations of the boundaries (shown in the high uncertainty in the radius) does account for this.

TABLE V

	Radius (R_E)	Centre (y,z) (R_E)	x-coordinate (R_E)
Fitted Magnetopause (MP)	26.6 ± 7	$(1.45, 2.50)$	-53.57 ± 6
Fitted Bow Shock (BS)	46.4 ± 9	$(1.02, -0.04)$	-37.09 ± 10
Weighted Average	N/a	$(1.38, 2.11)$	N/a
Literature MP [9]	26.3 ± 15	Assumed to be 0	-60

Table V. Circular fit parameters for the magnetopause (MP) and bow shock (BS) boundaries, derived from this study's classification methods. Reported values include boundary radii, centres in the $y-z$ plane, and mean x-coordinates. A weighted average of centre coordinates is included for reference. Values for the MP boundary from the literature are shown for comparison [9].

An analysis of the clock angles of all data points made outside the magnetotail indicates that the interplanetary magnetic field (IMF) has a positive B_z component, as shown in Fig. 22. Some sources suggest that this would cause the plasma sheet to become more distorted and shift northwards [43], [44]. This could explain the small positive z shift we see in the centres of the fitted MP.

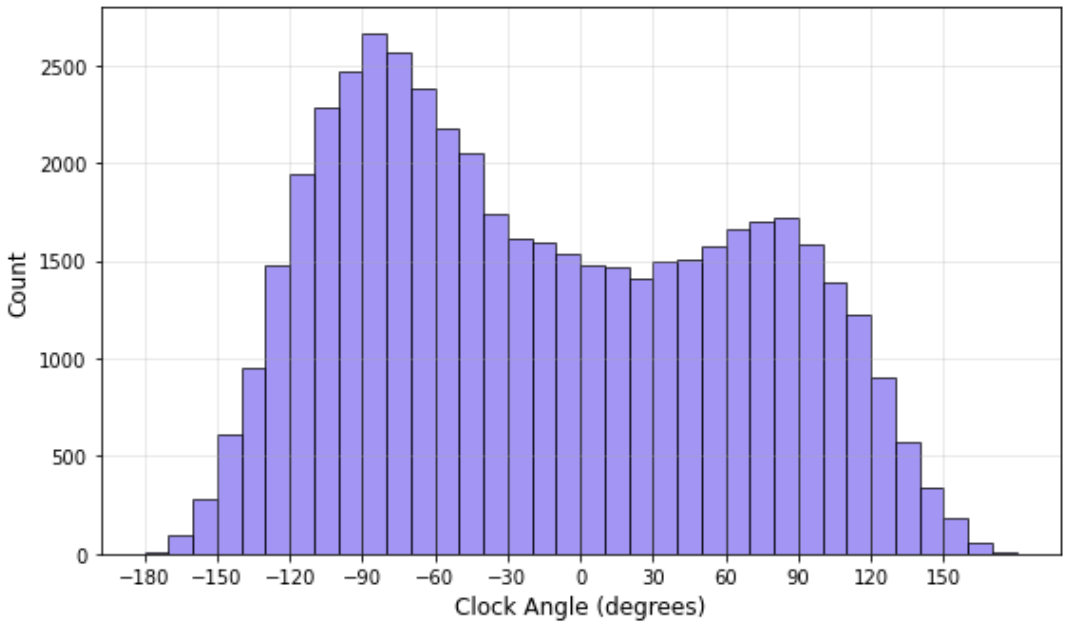


Figure 22. Histogram of hourly clock angle values of the magnetic field measured outside the magnetotail over the full dataset. The distribution is notably non-uniform, with a higher count in the positive z-direction relative to the negative z-direction. This may account for the slight positive z-offset observed in the fitted circular boundary models.

Similarly to how we labelled the MP boundary, we labelled the plasma sheet as the boundary between the NL and SL. In Fig. 23 we plotted a two-dimensional histogram of this boundary with a fitted mean z location for every y bin. As expected, this fit indicates that the plasma sheet is located on the $x - y$ plane with small fluctuations of ± 1 R_E.

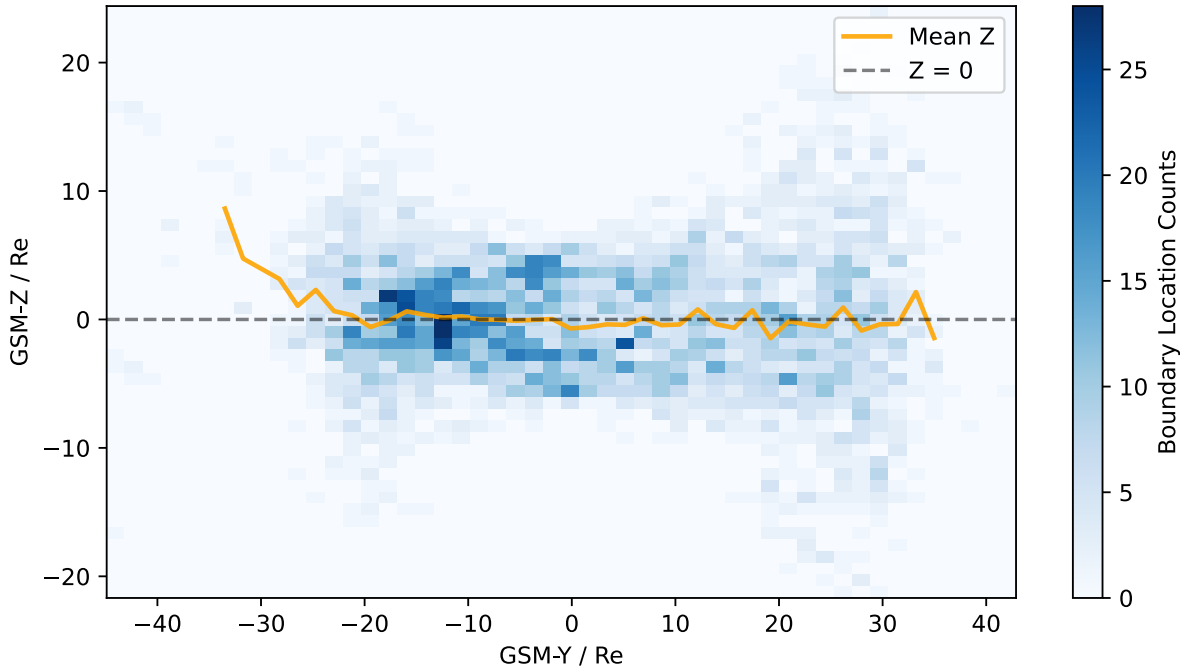


Figure 23. 2D histogram of plasma sheet boundary crossings - i.e., transitions between the north lobe and south lobe regions - projected in the geocentric solar magnetospheric $y - z$ plane. The colour intensity represents the number of boundary crossings per bin. The orange line shows the mean z-coordinate at each y-bin, and the dashed line marks the $z = 0$ reference.

3.2 Yearly Magnetosphere Plots – Solar Cycle

We plotted the magnetosphere cross-section for every year between 2011 and 2025. These fourteen plots are shown in Fig. 24. Solar maxima occurred in April 2014 and August 2024, and a solar minimum occurred in December 2019. To aid our comparative analysis, we overlapped years 2011, 2012, 2013, 2014 and 2015 with 2021, 2022, 2023, 2024 and 2025 respectively concerning the solar cycle, as shown in Fig. 24. There is no noticeable progression throughout the solar cycle regarding our fits. We see that the fitted centres of the paired years are closer to (0,0) than the centres of the single years, which is expected as the paired years include twice as many data points reducing the random uncertainty.

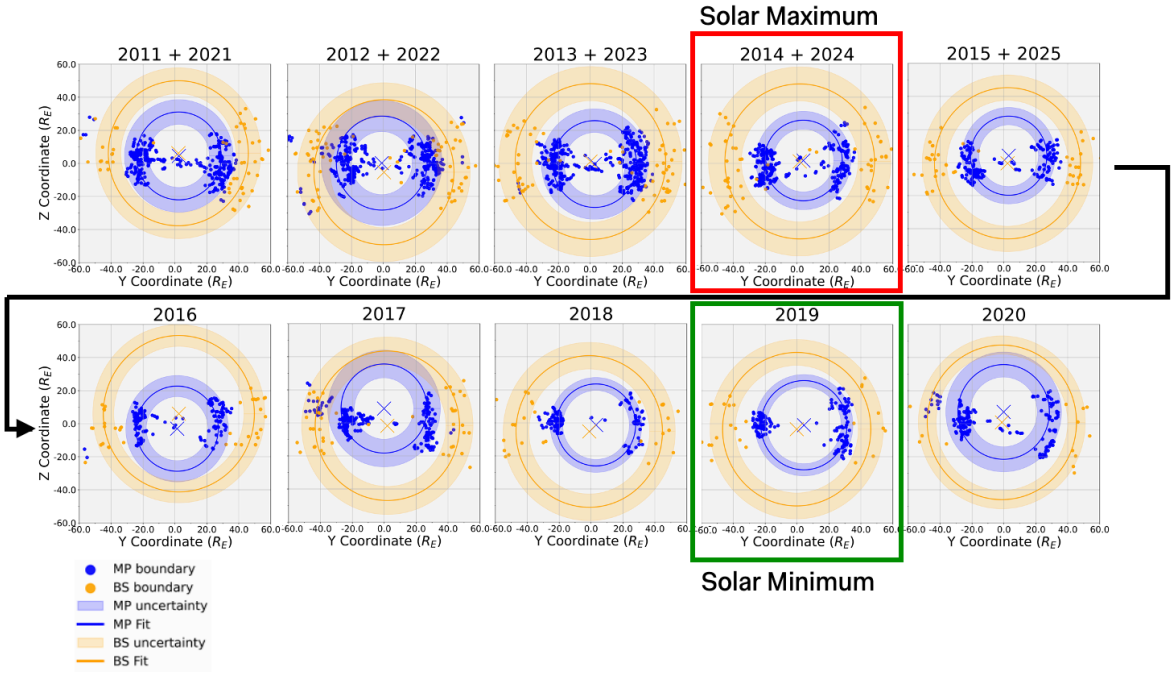


Figure 24. Circular fits of the magnetopause (MP) and bow shock (BS) boundaries in the geocentric solar magnetospheric $y - z$ plane, grouped by year pairs across the solar cycle. Years are paired based on solar cycle phase (e.g. 2011+2021), with solar maximum (red) and minimum (green) highlighted. Despite expectations, no clear trend in boundary structure is observed over the cycle.

As discussed in section 1.5, we were uncertain if the number of boundary crossings detected would correlate to the solar cycle. We argued that the increased frequency of solar events like CMEs during a solar maximum might result in greater dP_{dyn}/dt behaviour, which would cause the MP and BS boundaries to fluctuate at higher rates at lunar distance. In Fig. 25, we investigate the relationship between the number of boundary crossings and the solar cycle. We halved the number of crossings during paired solar cycle years. We see that the overall trend is shared across the MP and BS boundaries, with the MP crossings showing higher variation than the BS. The difference in variation is likely due to the different classification approaches used for each boundary. Nonetheless, there is no visible trend that follows the solar cycle maxima and minimum, with a peak in 2017 and a trough in 2015+2025. For future work, it would be interesting to investigate if any solar physical processes, such as dP_{dyn}/dt, relate to the number of MP and BS crossings detected by our classification processes.

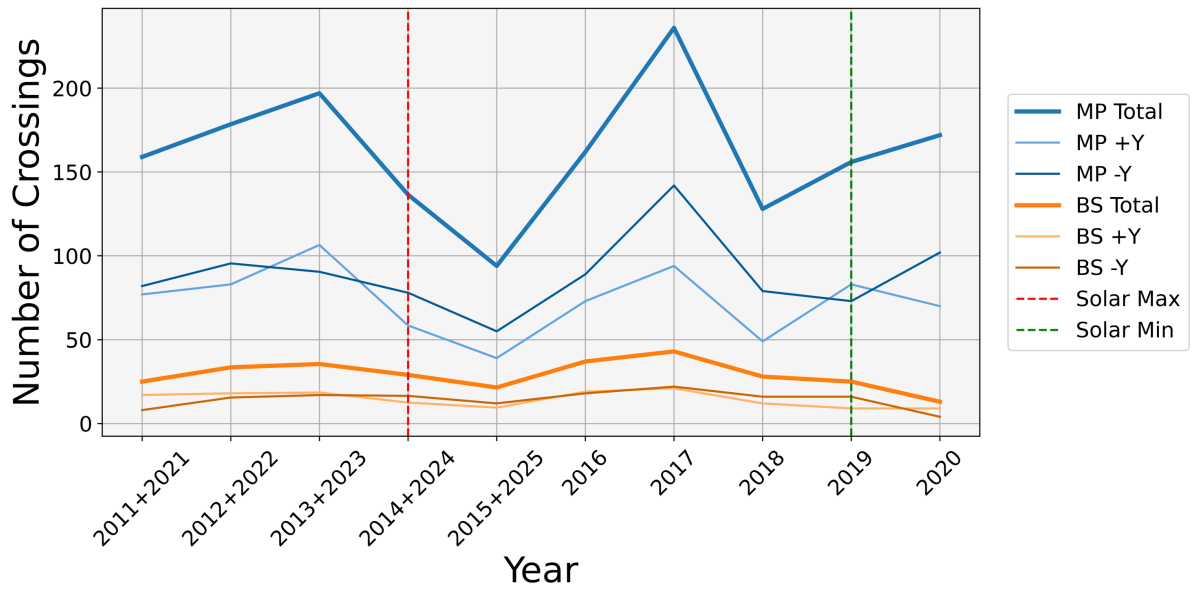


Figure 25. Yearly counts of magnetopause (MP) and bow shock (BS) crossings throughout the solar cycle, including directionality along the *y*-axis. Some years were paired to increase data coverage. A general solar cycle trend is not visible.

4 Future Work

In this section, we will discuss the extensions to this project which include building on the current model to improve its performance and further investigation of the solar cycle.

4.1 Two binary XGBoost Models

To tackle the higher rate of false positives in NL and SL in the XGBoost algorithm, due to the low proportion of data inside the magnetotail, we would test the performance of two consecutive binary XGBoost models. This would be analogous to the two binary *k*-means models used to label the data, still limiting it to only magnetic field data.

4.2 Sequential Nature of the Lunar Orbit

The Moon passes through the regions investigated in a set sequence due to the nature of the Moon's orbit (SW → MS → MT (NL or SL) → MS → SW). Implementing this into our classification process and/ or now-casting model would likely increase our overall accuracy.

4.3 Storm Index

Storm indexes are an attempt to quantise solar activity, their name relates to the geomagnetic storms caused in Earth's atmosphere by high solar activity. One of these storm indexes is called the Disturbance Storm Time (DST) index, which is calculated by measuring the magnetic field at low-latitude stations located on Earth. By importing the storm index for every existing data point, we can investigate its correlation to the distance of a boundary crossing to the circular fit of the boundary. If a correlation is found, we would have evidence suggesting that the fluctuations are caused by varying solar activity. This could act as another metric used to improve our classification process and/ or our now-casting model.

4.4 Aberration Correction

A similar approach was taken by Mieth et al. (2019) [7], where the solar wind velocity is taken from the NASA OMNI data set through OMNIWeb near the BS nose, which is the sunwards side of the bow shock. From this, we can calculate the induced aberration, which affected the data measured by THEMIS B & C downwind. By calculating the distance of the BS nose to THEMIS B & C, we would calculate the transit time of the incoming solar wind and apply the correction to the aberration using this delay. We would apply this approach and investigate any improvements in our model.

4.5 Now-casting the Bow Shock Boundary

After implementing the improvements above, we would aim to improve the BS now-casting classification process to improve the models' overall capabilities.

4.6 Increased Resolution

Furthermore, we would investigate the performance of our model at a higher resolution. This would decrease the response time of the now-casting model increasing its functionality. Due to the increased noise at higher resolution, there would likely be a compromise with the final accuracy of the model. We would investigate this compromise and allow users to make an informed decision depending on the application.

4.7 Further investigation of dP_{dyn}/dt

In addition, we would investigate the relationship of dP_{dyn}/dt with the solar cycle, with the number of MP and BS boundary crossings.

4.8 Publication and Gateway

Beyond publishing our findings, we would publish a comprehensive open-source model of our classification process and now-casting model and hopefully apply the model to Gateway once it is operational. Due to the unusual NHLO of Gateway, a more complete magnetosphere cross section would be mapped out, not only improving the now-casting model but also our understanding of the magnetosphere at lunar distance.

5 Conclusion

This project set out to build a now-casting model capable of classifying the magnetospheric region of a spacecraft at lunar distance using only magnetic field data. The aim was to support the deployment of the MAGIC magnetometer onboard the Lunar Gateway, by enabling the identification of the spacecraft's environment in real-time without the need for plasma measurements, which may be delayed or incomplete.

To achieve this, fourteen years of THEMIS B & C data were analysed and classified using a hybrid methodology: first applying unsupervised ML (k-means clustering) for data labelling, followed by training a supervised ML model (XGBoost) on the labelled dataset. The approach aimed at mitigating human bias produced interpretable clustering results and enabled a final model that achieved $96.2 \pm 1.1\%$ accuracy in identifying whether the spacecraft was in the SW, MT, NL or SL. This demonstrates that the project's central aim was achieved to a high standard.

The model's performance was validated both quantitatively via confusion matrices, precision-recall curves, and F1 scores and qualitatively through visual inspection of magnetospheric cross-sections consistent with established theory. Furthermore, the project provided insights into the magnetospheric dynamics across a solar cycle. Attempts to integrate a BS classifier revealed limitations in the current approach but highlighted promising directions for further development.

Although the main goals were achieved, several opportunities remain to extend this work. These include refining the now-casting model using a pair of binary classifiers, integrating physical constraints based on orbital sequencing, exploring correlations with solar storm indices, and applying aberration corrections. Most notably, future application of this model to Gateway's unique orbit could significantly enhance our understanding of the magnetosphere at lunar distance, fulfilling the project's long-term vision.

6 Acknowledgements

I would like to thank Professor Jonathan Eastwood for his invaluable guidance and support throughout this project. His weekly supervision sessions provided both direction and motivation, and his expertise was instrumental in shaping the scope and rigour of the work. I would also like to acknowledge my project partner for his dedicated work on the bow shock boundary classification, as well as for the collaborative support he offered throughout the project. His input and assistance were key in helping me progress with my section of the work.

7 Bibliography

[1] J. Cao, H. Wang, J. Zhang, and Q. Shi, 'Key Questions of Earth Wind–Moon Interaction', *Space Sci. Technol.*, vol. 4, p. 0094, Apr. 2024, doi: 10.34133/space.0094.

[2] R. Lanza and A. Meloni, Eds., 'The Earth's Magnetic Field', in *The Earth's Magnetism: An Introduction for Geologists*, Berlin, Heidelberg: Springer, 2006, pp. 1–66. doi: 10.1007/978-3-540-27980-8_1.

[3] B. T. Tsurutani *et al.*, 'Phase-steepened Alfvén waves, proton perpendicular energization and the creation of magnetic holes and magnetic decreases: The ponderomotive force', *Geophys. Res. Lett.*, vol. 29, no. 24, pp. 86-1-86–4, 2002, doi: 10.1029/2002GL015652.

[4] '(PDF) Modification of the Turbulence Properties at the Bow Shock: Statistical Results', ResearchGate. Accessed: Apr. 09, 2025. [Online]. Available: https://www.researchgate.net/publication/380547105_Modification_of_the_Turbulence_Properties_at_the_Bow_Shock_Statistical_Results

[5] N. A. Tsyganenko, 'A model of the near magnetosphere with a dawn-dusk asymmetry 1. Mathematical structure', *J. Geophys. Res. Space Phys.*, vol. 107, no. A8, p. SMP 12-1-SMP 12-15, 2002, doi: 10.1029/2001JA000219.

[6] J.-H. Shue *et al.*, 'A new functional form to study the solar wind control of the magnetopause size and shape', *J. Geophys. Res. Space Phys.*, vol. 102, no. A5, pp. 9497–9511, 1997, doi: 10.1029/97JA00196.

[7] J. Z. D. Mieth, D. Frühauff, and K.-H. Glassmeier, 'Statistical analysis of magnetopause crossings at lunar distances', *Ann. Geophys.*, vol. 37, no. 2, pp. 163–169, Mar. 2019, doi: 10.5194/angeo-37-163-2019.

[8] E. E. Woodfield, M. W. Dunlop, R. Holme, J. A. Davies, and M. A. Hapgood, 'A comparison of Cluster magnetic data with the Tsyganenko 2001 model', *J. Geophys. Res. Space Phys.*, vol. 112, no. A6, 2007, doi: 10.1029/2006JA012217.

[9] I. Gencturk Akay, Z. Kaymaz, and D. G. Sibeck, 'Magnetotail boundary crossings at lunar distances: ARTEMIS observations', *J. Atmospheric Sol.-Terr. Phys.*, vol. 182, pp. 45–60, Jan. 2019, doi: 10.1016/j.jastp.2018.11.002.

[10] ESA, 'Artemis introducing ERSA – European experiment to monitor radiation in deep space – ESA – Exploration'. Accessed: Sep. 27, 2024. [Online]. Available: https://blogs.esa.int/exploration/artemis-introducing-ersa-european-experiment-to-monitor-radiation-in-deep-space/

[11] NASA, 'Gateway Capabilities - NASA'. Accessed: Sep. 27, 2024. [Online]. Available: https://www.nasa.gov/gateway-capabilities/

[12] K. Ferrone, C. Willis, F. Guan, J. Ma, L. Peterson, and S. Kry, 'A Review of Magnetic Shielding Technology for Space Radiation', *Radiation*, vol. 3, no. 1, Art. no. 1, Mar. 2023, doi: 10.3390/radiation3010005.

[13] NASA, 'Gateway - NASA'. Accessed: Sep. 27, 2024. [Online]. Available: https://www.nasa.gov/mission/gateway/

[14] D. Connel and B. R. Zamora, 'A Lunar Orbit That's Just Right for the International Gateway - NASA'. Accessed: Sep. 27, 2024. [Online]. Available: https://www.nasa.gov/centers-and-facilities/johnson/lunar-near-rectilinear-halo-orbit-gateway/

[15] C. P. Newman, D. C. Davis, R. J. Whitley, J. R. Guinn, and M. S. Ryne, 'STATIONKEEPING, ORBIT DETERMINATION, AND ATTITUDE CONTROL FOR SPACECRAFT IN NEAR RECTILINEAR HALO ORBITS', 2018.

[16] 'ARTEMIS'. Accessed: Apr. 08, 2025. [Online]. Available: https://artemis.igpp.ucla.edu/orbits.shtml

[17] J. P. Eastwood, ‘Development of the ERSA-MAGIC instrument for magnetic field measurements on the Lunar Orbiting Platform Gateway’, European Space Weather Week (ESWW), Nov. 23, 2023.
[18] Imperial College London, ‘RadCube’, Imperial College London. Accessed: Sep. 27, 2024. [Online]. Available: https://www.imperial.ac.uk/a-z-research/space-and-atmospheric-physics/research/missions-and-projects/space-missions/radcube/
[19] D. Connel and K. Halloran, ‘Lunar Gateway Instruments to Improve Weather Forecasting for Artemis Astronauts - NASA’. Accessed: Sep. 27, 2024. [Online]. Available: https://www.nasa.gov/humans-in-space/lunar-gateway-instruments-to-improve-weather-forecasting-for-artemis-astronauts/
[20] ‘THEMIS’. Accessed: Apr. 19, 2025. [Online]. Available: https://themis.ssl.berkeley.edu/summary.php?year=2019&month=08&day=26&hour=0024&sumType=orb&type=sse_lunar
[21] ‘THEMIS PDMP Draft 20041018’.
[22] M. E. Innocenti, J. Amaya, J. Raeder, R. Dupuis, B. Ferdousi, and G. Lapenta, ‘Unsupervised classification of simulated magnetospheric regions’, *Ann. Geophys.*, vol. 39, no. 5, pp. 861–881, Oct. 2021, doi: 10.5194/angeo-39-861-2021.
[23] E. Camporeale, ‘The Challenge of Machine Learning in Space Weather: Nowcasting and Forecasting’, *Space Weather*, vol. 17, no. 8, pp. 1166–1207, 2019, doi: 10.1029/2018SW002061.
[24] M. Ahmed, R. Seraj, and S. M. S. Islam, ‘The k-means Algorithm: A Comprehensive Survey and Performance Evaluation’, *Electronics*, vol. 9, no. 8, Art. no. 8, Aug. 2020, doi: 10.3390/electronics9081295.
[25] B. Bochenek and Z. Ustrnul, ‘Machine Learning in Weather Prediction and Climate Analyses—Applications and Perspectives’, *Atmosphere*, vol. 13, no. 2, Art. no. 2, Feb. 2022, doi: 10.3390/atmos13020180.
[26] S. Chakraborty, N. K. Nagwani, and L. Dey, ‘Weather Forecasting using Incremental K-means Clustering’, 2014.
[27] T. Colak and R. Qahwaji, ‘Automated Solar Activity Prediction: A hybrid computer platform using machine learning and solar imaging for automated prediction of solar flares’, *Space Weather*, vol. 7, p. 12, Jun. 2009, doi: 10.1029/2008SW000401.
[28] E. Camporeale, A. Carè, and J. E. Borovsky, ‘Classification of Solar Wind With Machine Learning’, *J. Geophys. Res. Space Phys.*, vol. 122, no. 11, p. 10,910-10,920, 2017, doi: 10.1002/2017JA024383.
[29] J. T. Page, Z. S. Liechty, M. D. Huynh, and J. A. Udall, ‘BamBam: genome sequence analysis tools for biologists’, *BMC Res. Notes*, vol. 7, no. 1, p. 829, Nov. 2014, doi: 10.1186/1756-0500-7-829.
[30] A. M. Ikotun, A. E. Ezugwu, L. Abualigah, B. Abuhaija, and J. Heming, ‘K-means clustering algorithms: A comprehensive review, variants analysis, and advances in the era of big data’, *Inf. Sci.*, vol. 622, pp. 178–210, Apr. 2023, doi: 10.1016/j.ins.2022.11.139.
[31] A. Shahcheraghian, A. Ilinca, and N. Sommerfeldt, ‘K-means and agglomerative clustering for source-load mapping in distributed district heating planning’, *Energy Convers. Manag. X*, vol. 25, p. 100860, Jan. 2025, doi: 10.1016/j.ecmx.2024.100860.
[32] A. Pal, ‘Gradient Boosting Trees for Classification: A Beginner’s Guide’, Medium. Accessed: Apr. 16, 2025. [Online]. Available: https://medium.com/swlh/gradient-boosting-trees-for-classification-a-beginners-guide-596b594a14ea
[33] ‘What is XGBoost?’, NVIDIA Data Science Glossary. Accessed: Apr. 05, 2025. [Online]. Available: https://www.nvidia.com/en-gb/glossary/xgboost/
[34] T. Chen and C. Guestrin, ‘XGBoost: A Scalable Tree Boosting System’, in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data*

Mining, in KDD '16. New York, NY, USA: Association for Computing Machinery, Aug. 2016, pp. 785–794. doi: 10.1145/2939672.2939785.
[35] 'f1_score', scikit-learn. Accessed: Apr. 10, 2025. [Online]. Available: https://scikit-learn/stable/modules/generated/sklearn.metrics.f1_score.html
[36] P. Charbonneau, 'Dynamo models of the solar cycle', *Living Rev. Sol. Phys.*, vol. 17, no. 1, p. 4, Jun. 2020, doi: 10.1007/s41116-020-00025-6.
[37] T. Pulkkinen, 'Space Weather: Terrestrial Perspective', *Living Rev. Sol. Phys.*, vol. 4, no. 1, p. 1, May 2007, doi: 10.12942/lrsp-2007-1.
[38] D. H. Hathaway, 'The Solar Cycle', *Living Rev. Sol. Phys.*, vol. 12, no. 1, p. 4, Sep. 2015, doi: 10.1007/lrsp-2015-4.
[39] A. A. Samsonov, Y. V. Bogdanova, G. Branduardi-Raymont, J. Safrankova, Z. Nemecek, and J.-S. Park, 'Long-Term Variations in Solar Wind Parameters, Magnetopause Location, and Geomagnetic Activity Over the Last Five Solar Cycles', *J. Geophys. Res. Space Phys.*, vol. 124, no. 6, pp. 4049–4063, 2019, doi: 10.1029/2018JA026355.
[40] N. Lugaz, C. J. Farrugia, W. B. Manchester IV, and N. Schwadron, 'THE INTERACTION OF TWO CORONAL MASS EJECTIONS: INFLUENCE OF RELATIVE ORIENTATION', *Astrophys. J.*, vol. 778, no. 1, p. 20, Oct. 2013, doi: 10.1088/0004-637X/778/1/20.
[41] W. S. Shang *et al.*, 'Unusual Location of the Geotail Magnetopause Near Lunar Orbit: A Case Study', *J. Geophys. Res. Space Phys.*, vol. 125, no. 4, p. e2019JA027401, 2020, doi: 10.1029/2019JA027401.
[42] 'THEMIS'. Accessed: Apr. 20, 2025. [Online]. Available: https://themis.ssl.berkeley.edu/summary.php?year=2010&month=09&day=20&hour=0024&sumType=thb&type=overview
[43] T. C. Liu *et al.*, 'IMF By influence on fast earthward convection flows in the near-lunar magnetotail', *Geosci. Lett.*, vol. 12, no. 1, p. 6, Feb. 2025, doi: 10.1186/s40562-025-00379-5.
[44] K. Maezawa *et al.*, 'Structure of the distant magnetotail and its dependence on the IMF By component: GEOTAIL observations', *Adv. Space Res.*, vol. 20, no. 4, pp. 949–959, Jan. 1997, doi: 10.1016/S0273-1177(97)00503-6.